Recall and Response:
Relationship Adjustments to Adverse Information Shocks

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Abstract

How resilient are U.S. buyer-foreign supplier relationships to new information about product defects? We construct a novel dataset of U.S. consumer-product recalls sourced from foreign suppliers between 1995 and 2013. Using an event-study approach, we find that compared to control relationships, buyers that experience recalls temporarily reduce their probability of trading with the suppliers of the recalled products by 17%. The reduction is much larger for new than established buyer-supplier relationships. Buyers that experience a recall are more likely to add other suppliers to their portfolios, diversifying supplier-specific risk in the aftermath of a recall; this effect, too, is larger for buyers impacted by recalls in new relationships. There is a long lag -- up to two years -- before diversification, consistent with a high cost of establishing new relationships.

Keyword*:* Buyer-supplier relationships; information flows; recalls; consumer products

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1 Introduction

Relationships between individual buyers and suppliers are the building blocks of international trade. Understanding how these relationships adjust in response to new information is fundamental to understanding how global production networks function. We use a quasi-random experiment setting to explore the margins of adjustment for buyer-supplier relationships that experience negative information shocks. Specifically, we investigate the impacts of consumer-product recalls for health or safety reasons on the relationships between U.S. firms and their foreign suppliers.

Supplier matching and switching costs are especially salient in cross-border transactions due to the pervasiveness of imperfect information and heterogeneous contract enforcement across countries (Startz, 2018; Macchiavello and Morjaria, 2015; Bernard, Moxnes, and Ulltveit-Moe, 2018; Monarch, forthcoming). *A priori*, it is not obvious to what extent a buyer-supplier relationship would respond to a negative information shock at the relationship level. If the buying firm perceives the product defect precipitating a recall to be due to a manufacturing error or miscommunication, the shock may strain the relationship. The recall event could also prompt the supplier to contract operations. Conversely, if the buyer believes that it is responsible for the defect, perhaps because it provided faulty or incomplete designs, it may continue the relationship. A further complicating factor is that buyers and suppliers write complex contracts that specify monetary penalties for suppliers under certain conditions.

Consumer-product recalls are specific to a buyer (typically, a retailer or distributor), supplier (manufacturer), and product, and hence serve to identify information shocks at the relationship-product level. The recall process begins as a result of internal quality-control checks or complaints submitted directly to the seller or government by consumers, competitors, or health officials. A U.S. firm whose product is under investigation receives an official notice of the investigation and is apprised of the conclusions before the public recall announcement. The process conveys information to a distributor or retailer about the

quality of the production process at the supplying firm.

To study the response to these shocks, we construct a novel dataset of the universe of U.S. consumer-product recalls sourced from foreign suppliers between 1995 and 2013 from Consumer Product Safety Commission (CPSC) public notices. We identify the imported shipments most likely associated with a recall in confidential customs transaction records linked to firm identifiers by the U.S. Census Bureau, using information about the U.S. buyer, foreign supplier, product, and shipment date. We also use the customs transaction data to identify control relationships: buyer-supplier pairs with similar trade patterns, trading similar products, around the same time as the treated pairs.

Once we identify buyer-supplier relationships impacted by recalls, along with appropriate control relationships, we use the high frequency of the customs records to trace the precise timing of adjustments firms make both in anticipation of and immediately following the recall announcement. Tracing the firms' transactions in the period before and after the recall announcement enables us to identify the timing of the information shock and infer which parties knew about the product defect before it was announced and how long before, shedding light on how information is transmitted in supply networks.[1]

We implement an event-study approach that treats the recall process as a quasi-random experiment. Although the types of product defects that trigger recalls may be precipitated by the firms' choices and actions, recalls are both rare and, empirically, hard to predict. Nevertheless, our empirical specifications include buyer-supplier fixed effects that control for the possibility that some buyer-supplier pairs have a higher probability of experiencing a recall than others.

We begin by establishing two key results that are consistent with our view of recalls as quasi-random. First, compared to control relationships, relationships shocked by a product

[1]Our interest in the shock to *firms'* information sets distinguishes our paper from papers that analyze the impact of recalls through the lens of *consumer* response, such as Freedman, Kearney, and Lederman (2012) and Zhong (2018).

defect experience a seven-percentage-point decrease in the probability of trade that persists for two quarters. Second, this sharp drop – equivalent to a decline in the probability of trade of about 17% – begins, on average, two quarters *before* the recall is publicly announced, consistent with the announcement being a symptom, and not the cause, of the disruption. Combined, these findings suggest that the product recall – or, more precisely, the information shock that precedes it – is unanticipated by the trading firms and is addressed promptly.

The magnitude of a recall's effect on trade depends on the age of the trading relationship. The effect on new relationships, which we define as those that had not traded in the second or third year prior to the recall, is a decrease of 15 percentage points – three times the size of the effect on established buyer-supplier relationships. This finding is consistent with the idea that a U.S. buyer extracts some information about the quality of a foreign supplier from a recall. For long-term relationships, there is not much updating, but for new relationships this information is quite disruptive. The finding is also consistent with the possibility that recalls in established relationships, in which both parties are heavily invested, are easier to fix due to longstanding communication channels. Although the estimates are not as precise, new relationships also have a lower probability of resuming trade for as long as three years after the recall announcement.

Product recalls have both direct and indirect effects on trading relationships. The decline in the probability of trade is not limited to the recalled product, but impacts all products sourced from the same foreign supplier. The probability of trading any other products with the same supplier drops by six percentage points, or about 12%, in the short run. The effect in the long run is less precisely identified, but point estimates suggest a long-lasting, if muted, negative effect on the relationship.

A recall indirectly affects both the buyer's and the seller's other trade relationships as well. On the buyer side, we see evidence of a long-run increase in reliance on other suppliers, particularly for the same product and especially from countries other than the impacted foreign supplier's country. The probability that the U.S. buyer buys the same product from

a supplier in a different country increases by about ten percentage points, or 20%, in the long run. On the supplier side, we find suggestive evidence that the supplier loses orders from other U.S. buyers. The probability that a supplier experiencing a recall trades with a different U.S. buyer falls, compared to control suppliers, by about five percentage points in the long run, a decrease of about 10% from the baseline probability.

Our paper provides, to the best of our knowledge, the first analysis of how buyers' and sellers' relationships adjust following relationship-specific negative shocks. This is especially relevant in an era where goods production is increasingly fragmented across national borders (see, e.g., Fort, 2017; Antrás, Fort, and Tintlenot, 2017). Our findings that supplier diversification takes several quarters suggest that trade relationships take time to develop. Even in the context of consumer products (e.g., apparel, footwear, leather goods), where relationship-specific investments are relatively low and there are many possible alternative suppliers, our finding that established relationships are less likley to discontinue compared to new relationships underscores the value of long-term relationships in international trade.

Our paper examines relationship-specific shocks. Other papers about the propagation of shocks in production networks have typically examined large-scale shocks such as natural disasters. For example, Barrot and Sauvagnat (2016, Figure IV) find that a firm's sales growth falls substantially when a natural disaster (e.g., a blizzard or earthquakes) affects a major supplier to that firm. The decline takes 3–4 quarters to materialize and does not return to baseline for ten quarters. In the context of our study, the foreign suppliers remain available and accessible; there are no physical barriers such as power disruptions or damaged infrastructure that prevent the buyer from continuing to contract with the supplier, so we observe the buyer's unconstrained trading decision. Boehm, Flaaen, and Pandalai-Nayar (2019) study how the March 2011 Tohoku earthquake disrupted production of Japanese multinational parent firms supplying key inputs to their U.S. affiliates. They find that imports by U.S. affiliates fell immediately in the month following the earthquake and did not fully recover for six months. Our finding that trading probability returns to baseline

after only two quarters highlights that the nature of the firm-level shock impacts the speed and duration of its impact.

The rest of the paper is organized as follows. Section 2 provides background on the recall process. Section 3 describes our data sources and our procedure for merging them, identifying recalled shipments and affected relationships, and identifying appropriate control shipments and relationships. We lay out the empirical approach and results in Section 4. Section 5 concludes.

2 Product Recalls in the United States

Product recalls represent a breakdown in the production process, due to design, communication, or manufacturing errors (Lyles, Flynn, and Frohlich, 2008). Our study uses information on consumer-product recalls issued by the Consumer Product Safety Commission, which was established by the 1972 Consumer Product Safety Act. The CPSC promotes the safety of over 15,000 consumer goods such as toys, furniture, consumer electronics, and all-terrain vehicles.[2] Product recalls are conducted as part of CPSC's goal of addressing risks of injury to consumers. The main objectives of a recall are identifying and locating all defective products, removing them from the distribution chain and possession of consumers, and communicating information to the public about the defect, hazard, and corrective action (U.S. Consumer Product Safety Commission Office of Compliance & Field Operations, 2012). Figure 1 shows the number of recall announcements of imported products, by year, between 1995 and 2013. The peak in 2007 represents the wave of recalls of toys manufactured in China.

There are at least four types of costs to firms associated with product failures. First, there is the cost of the investigation into the cause of the problem, which may disrupt production and promotion schedules. Second, there is the cost of issuing refunds or replacement

[2]The main consumer products excluded from CPSC's jurisdiction are automobiles, guns, food, cosmetics, and drugs.

products or providing repairs. Third, there is the cost of addressing defects in the product design, production process, material sourcing, etc. Finally, there may be a cost of lost consumer goodwill, which can last for a long time (Kumar and Schmitz, 2011; Zhong, 2018). For all these reasons, companies have an incentive to minimize their risk of product failures.

Product recalls are a process that culminates with the posting of a public notice in the form of a press release: the recall announcement. The process begins when a complaint is filed with CPSC or the company. The complaint may originate with a consumer, a competitor, or a professional such as a healthcare provider who believes the product may have caused or poses harm. The complaint may also come from the manufacturer, importer, distributor, or seller of the product if the firm discovers a defect or is notified of one by a customer. Companies that have been notified of a possible hazard related to their products have a legal duty to report this information to CPSC within 24 hours (U.S. Consumer Product Safety Commission Office of Compliance & Field Operations, 2012, pp. 6–8). CPSC then launches an investigation to identify the product hazard using four main criteria: pattern of defect, number of defective products in commerce, severity of risk, and likelihood of injury. In some cases, the investigation concludes that no corrective action is needed, and the case is closed. If CPSC deems that a product poses a risk to consumers, it initiates a recall by sending a letter to the company. The company must then design a corrective-action plan, which may entail ascertaining the nature and scope of the defect, discontinuing production, ordering replacements, halting retail sales, devising a communications and consumer-response plan, or installing upgrades to quality-control systems to avoid future hazards.

Figure 2 depicts a stylized timeline for a recall, in which the product is no longer sold to consumers when the firm becomes aware of a problem in production. If, instead, the product is still available for sale at the time of the information shock, the firm may discontinue sales then, or wait for the results of the investigation. Only once a corrective-action plan is in place is the public notified of the recall.

This complex process implies that by the time the press release is made available to the

public – i.e., the product is officially recalled – the buyer, and likely its supplier, have had the opportunity to prepare a response to the recall. Unfortunately, we do not observe the exact date at which the firms first discover, or begin to address, the product or process defect. Instead, we have the date of the recall announcement; the dates of any shipments we have associated with the recall announcement; and, in most cases, also the dates of production and/or sale, as reported in the recall announcement. On average, in our sample of recalls, sale to consumers begins within a couple of weeks of the earliest production date (p_1 to s_1 in Figure 2) and continues for several months after the last production date (p_2 to s_2). On average, a few months separate the last date of sale from the date of the recall announcement (s_2 to r). The lag between last sale and recall announcement can range from zero (sales cease with recall announcement) to several years.[3] Because we cannot identify the exact date when the buying firm first becomes aware of an issue we treat the recall announcement itself as an outcome of the unobserved negative shock to the buyer-supplier-product tuple.

3 Data

3.1 Data Sources

Our empirical analysis relies on three sources of administrative data: import transactions data from the Longitudinal Firm Trade Transactions Database (LFTTD), recall data from CPSC, and business name from the Business Register (BR).

[3]There is considerable heterogeneity in the length of the recall process. In the later years of our sample, an increasing share of recalls were handled under CPSC's "Fast-Track" program, which allows a recall to take place without a full investigation; such recalls are typically completed within 40 business days. Unfortunately, recall notices did not begin specifying the "Fast-Track" status of a recall until 2013. Therefore, for effectively all our sample, we cannot distinguish "Fast-Track" recalls. In other cases, recalls may take a longer time. Recalls of certain products require complex planning, for example in the case of batteries, where disposal of the item can itself be hazardous. Very large recalls can also require extra planning. In other cases, products may cross regulatory jurisdictions. For instance, if an injury is caused by the straw in a drinking cup then both CPSC and the Food and Drug Administration have jurisdiction over the case, and the agencies must agree on the details of the recall.

3.1.1 Longitudinal Firm Trade Transactions Database and Business Register

The LFTTD contains transaction-level detail on the universe of imported shipments valued over US$2,000 of merchandise into the U.S. starting in 1992 (Kamal and Ouyang, 2020). These data originate with U.S. Customs and Border Protection (CBP) and are augmented with a firm identifier by the U.S. Census Bureau. There are seven variables key to our analysis: an identifier for the buyer (U.S. firm), an identifier for the manufacturer or supplier (foreign firm), the date of the transaction, the product traded (HS10), the shipment's country of origin, the dollar value of the shipment, and whether or not the supplier and buyer are affiliated.[4]

The supplier's identity is encoded in an alphanumeric variable, called the manufacturer identifier (MID), which includes a two-digit country code, the first three letters of the first word and the first three letters of the second word in the supplier's name, the numeric portion of the street address, and the first three letters of the city name.[5] We remove the numeric and city portion of the variable to create a supplier-country pair identifying the supplier. Within a country, the abbreviated supplier name enables us to link shipments to the recall data. We describe the linking process in Appendix A.1 and discuss the aggregation of MIDs in Appendix A.3.

The BR is a continuously updated database of employer business establishments operating in the U.S. and includes business names at the establishment level and a unique firm identifier indicating common ownership of a given set of establishments (DeSalvo, Limehouse, and Klimek, 2016). We match the U.S. buyer names listed in the publicly available product-recall announcements to the universe of business names in the BR to assign a firm

[4]Affiliated relationships are those in which one firm owns a stake of at least 6% in the other. It is an indicator variable; we do not observe the actual equity shares. The ownership relationship of trading parties is denoted by item 32.C in Form-7501.

[5]For example, Basker-Kamal Enterprises, 4600 Silver Hill Road, Washington, DC 20233, USA, would have an MID: US-BASKAM-4600-WAS. CBP requires that the MID constitute the manufacturer, not trading companies or other intermediaries. See Kamal and Monarch (2018) for more details on the manufacturer identification code.

identifier to a recall announcement, which can then be linked to the LFTTD.

3.1.2 Recall Data

The CPSC data used in this study include all consumer-product recalls from 1995 to 2013. The dataset includes over 5,000 recall notices, of which approximately 3,000 are associated with imported products.

Most recall notices include only a single product, but some include several products that may or may not share similar characteristics. The vast majority – 90% – of the notices are dated after 2001.[6] We focus on creating consistent variables to enable linking the recall data to the LFTTD. Figure 3 shows an example of an official press release issued by CPSC.[7] We extract, from each press release, the date of the recall announcement, the name(s) of the U.S. buyers (the firms importing, distributing, or retailing the product), the product type and detailed product description, the name(s) and countries of the foreign suppliers, and the dates or date ranges of manufacture, import, and sale. Not all these variables are available for every recall. We exclude recalls with missing key information, such as the name of the U.S. buyer. We use the product name and detailed description to assign each recall one or more HS codes.[8]

To get a sense of how prevalent recall events are across broad product categories, the first column of Table 1 shows the count of recalls of imported products for each of the top nine HS2 categories in the CPSC database, which collectively represent more than three quarters of the recalls of imported products. In the second column, we calculate the percentage of import shipments over the 1995–2013 period that are associated with a recall.[9] Recalls are

[6] CPSC occasionally re-issues press releases, e.g., when the injuries are severe or there are changes to the remedy offered. Our dataset excludes such updated releases.

[7] This example is given for illustrative purposes only. The recall was announced in 2017, four years after the end of our sample period.

[8] We assign each product description within a recall notice at least a HS4 code, adding detailed HS codes as the descriptions permit.

[9] A product recall may be associated with multiple products. In these cases, we count a recall multiple times across product categories.

extremely rare events. Among all product categories, toys are most likely to be recalled: 947 recalls include one or more toys. As a share of shipments, however, toy recalls account for only 0.01%. In the last column, we report the number of unique recalls that we link to the LFTTD as a share of the total number of recalls in that HS2 category. These range from a match rate of less than 30% for articles of plastic and for apparel (not knitted or crocheted) to nearly 50% for furniture and home furnishings.

3.2 Treated and Control Shipments

We link the product-recall data, which contain the U.S. buyer name, foreign supplier name, country of origin, product code assigned by us based on the product description, and a set of dates (production, import, sale, and recall announcement), to the LFTTD in order to identify shipments associated with the recall. We start by linking the importer names obtained from the recall announcements to business names at the establishment level to assign one or more possible firm identifiers to each shipment, and narrow down the possible matches using information about the product, country of origin, supplier name, and dates. The linking process is described in detail in Appendix A.1.

We find one or more shipments that match the available information on the recall notice for approximately 43% of the notices. These are our **treated shipments**. The treated shipments are dominated by imports of furniture, toys, electrical machinery, clothing, and plastic products, sourced primarily from China, Hong Kong, Taiwan, Canada, Mexico, and Japan.

There are hundreds of millions of shipments in the LFTTD, so it is computationally impractical to include all shipments, or even shipments matching the recalled shipments' HS4 or HS6 codes, in the analysis. Instead, we use the following procedure to select **control shipments** for each treated shipment. We start by identifying shipments that share a country of origin, a broad product classification (HS4), and a similar time period (within 90 days of the recalled shipment). We refine the control set by matching the treatment's pre-

recall firm- and relationship-level characteristics including the firm's total import value over a three-year period and the relationship-specific trade value over the same period. Recalled shipments for which we cannot assign controls are dropped from our analysis.

We take several steps to ensure that our control observations are not impacted by the recall either directly or indirectly. The recalls we identify are generally specific to the relationship and product, unlike recalls that are industry-wide (e.g., recalls of airbags by Takata that impacted many automobile manufactures; see Consumer Reports, 2018), so importers and exporters of similar products should not be directly impacted by a recall. To ensure no indirect effects, we make two additional restrictions on the set of controls. First, although the control shipments share an HS4 product classification with the treated shipments, we exclude any controls that share an HS6 product classification with the treatment. Second, we exclude from our set of control buyers any buyer that had ever traded with the treated supplier; and likewise exclude any control supplier that had ever traded with the treated buyer. These conditions are relaxed in robustness checks reported in Appendix B.2. Finally, firms involved in any other recalls are excluded from the control group.

3.3 Analysis Sample

Using the control shipments described above, we construct two samples for our analysis. The first is the "buyer sample." This sample restricts the control shipments to those whose buyers and buyer-seller pairs match the characteristics of the treated buyers and buyer-seller pairs. Specifically, we identify buyers with similar trading values over the three years prior to the recall announcement, and pairs with the same affiliation status, similar value of trade over the three-year period, and similar share of the traded product in their portfolios. We describe the matching process in more detail in Appendix A.2. The second sample is the "supplier

sample," which restricts controls based on supplier, rather than buyer, characteristics.[10]

To study the impact of product recalls, we follow the trading history of the buyers and suppliers associated with each treated and control shipment. We divide the 1,080 days before and after each recall into 90-day intervals, with 12 pre-recall quarters and 12 post-recall quarters. We use this constructed quarterly dataset to study the evolution of trading relationships at a relatively high-frequency, granular level. For each pair of firms, consisting of a buyer b and a supplier s, in each quarter t, we determine whether, and how much, the pair traded the recalled product p as well as other products; whether the buyer traded with other foreign suppliers (from the same country or from other countries); and whether the supplier traded with other U.S. buyers. We restrict all U.S. buyers in our sample to those with at least one shipment (of any good, from any supplier) in each of the 16 quarters closest to the recall, which removes any exiters from our sample. This allows us to examine trading behavior conditional on the firm continuing to exist and trade.[11] The analysis samples include approximately 750 treated and 4,200 control relationships in each quarter.

Panel A of Table 2 provides summary statistics on the treated and control buyers in the buyer sample, using data from quarters (-12) through (-5) relative to the recall announcement.[12] For the treated relationships, across the eight quarters, the average probability that the buyer purchases the recalled HS6 product from the supplier is 41%; the corresponding number for control relationships is 38%. The average quarterly probability of the buyer importing one or more shipments of some other product (different HS6) from the same supplier is 52% for treated firms and 51% for control firms. The average quarterly probability of the

[10]The buyer and the supplier samples differ not only in the controls but also in the treated observations. Treated relationships in the buyer sample may not be in the supplier sample if we cannot find suitable controls based on supplier charateristics.

[11]In practice, this removes a very small number of firms from our sample. A robustness check in Appendix B.2 removes this restriction.

[12]Guided by conversations with CPSC officials, our understanding is that the average recall process (including investigation and remedial action) lasts between several weeks and several months, but typically less than a year. We therefore consider the second and third years prior to the public announcement to be a period most likely uncontaminated by the recall process.

buyer importing the recalled HS6 product from one or more different suppliers is about 62% for treated firms and 69% for control firms. We also report the average probability that the buyer imports the recalled HS6 product from other suppliers located in the same country as the recalled supplier and the average probability that the buyer imports the recalled HS6 product from other countries; both treated and control firms are more likely to purchase the recalled HS6 product from suppliers located in the same country than in other countries. Both treated and control firms have a very high probability (roughly 97%) of purchasing other products from other suppliers. Finally, treated firms' average quarterly value of imports purchased from the supplier is about $3.2 million (in 2015 dollars); an additional $100 million worth of imports are purchased from other suppliers. For buyers in the control group, the average value of imports purchased from the supplier of the control shipment is about $7 million, with about $60 million purchased from other suppliers. The average traded values are influenced by the presence of very large importers in both samples. Therefore, we also report the median traded values; these are similar across the treated and control groups for both variables.

Panel B of Table 2 shows statistics for the supplier sample. That the probability of the supplier selling the HS6 product to the buyer in any given quarter (41% for treated suppliers, 39% for controls) is numerically almost identical to the probability, from Panel A, of the buyer buying the HS6 product from the seller indicates the similarity of the two samples. That the probability of the supplier selling the HS6 product to a different buyer per quarter (50% for treated suppliers, 47% for controls) is also very similar to the buyer sample is a coincidence.

Overall, the key variables in the treatment and control groups two years prior to the recall announcement are very similar. Our event-study results in the next section further demonstrate that the treated- and control-group responses exhibit parallel trends in the pre-shock period.

4 Results

4.1 Regression Specification

We begin by estimating the following regression equation using data from the six-year interval surrounding the recall announcement:

$$\text{Import}_{bspt} = \sum_{\tau=-12}^{+11} \pi_\tau \text{Quarter}_{t+\tau} + \gamma_{bs} + \sum_{\tau=-12}^{+11} \beta_\tau \text{Treated}_{bs} \text{Quarter}_{t+\tau} + \varepsilon_{bspt} \tag{1}$$

where Import_{bspt} is an indicator for buyer b having one or more shipments of product p (defined at the HS6 classification level) from supplier s in quarter t.[13] The relationship-specific fixed effect, $\boldsymbol{\gamma}_{bs}$, captures unobservable differences in the propensity to import at the buyer-supplier level in the window of time surrounding the recall.[14] $\text{Quarter}_{t+\tau}$ is a quarter fixed effect, where $\tau = 0$ indicates the quarter of the recall announcement. Treated_{bs} is a time-invariant indicator that equals one if buyer b experienced a recall of a product purchased from supplier s. On its own, Treated_{bs} is perfectly collinear with the relationship fixed effects, but interacted with the quarter fixed effects, it allows us to identify the differential trade patterns for treated and control relationships. Standard errors are clustered by recall, allowing for arbitrary autocorrelation in the error terms for a given treatment or control observation and for arbitrary correlation across treatment and control observations for the same recall. We estimate similar equations using other outcome variables, such as whether the buyer imported other products from the same supplier, or whether the supplier sold the same product to other buyers.

The vector of parameter estimates $\widehat{\boldsymbol{\beta}}$ describes how the trading relationship evolves before and after the recall announcement. As noted earlier, the information shock to the

[13] Most of the adjustment in our data takes place on the extensive margin. For completeness, in Appendix B.3.1, we report results from a specification using the log of import value to study the intensive margin of trade.

[14] If a single buyer-supplier-product tuple experiences more than one recall, we allow separate fixed effects associated with the observations surrounding each recall.

buyer-supplier relationship predates the announcement, possibly by a long period. We therefore start the event-study window 12 quarters prior to the recall announcement, and view both the announcement and the buyer-supplier relationship in the months before as well as after the announcement as outcomes of this unobserved shock. The designation $\tau = 0$, for the quarter immediately following the announcement, is a convenient normalization rather than an indication of the date of the shock.

A causal interpretation of the coefficients requires that the assignment of treatment and control buyer-supplier pairs be independent of the error term. In our context, the assignment of treatment – pairs that experience a recall – may not be completely random. Some products, countries, or firms may have a higher propensity than others to experience recalls. Our key identifying assumption is that recalls are uncorrelated with time-varying factors that directly influence trade. Although this assumption is not directly testable, in Appendix B.1 we provide suggestive evidence that recalls are largely unpredictable.

4.2 Relationship-Level Effects

Figure 4 shows the estimated coefficient vector $\widehat{\boldsymbol{\beta}}$, along with 95% confidence intervals, from Equation (1). The dependent variable is $\mathbb{I}\left(\text{Imports}_{bspt} > 0\right)$, an indicator that equals 1 iff buyer b purchases recalled product p from supplier s in quarter t. To interpret the 24 coefficients, note that the recall announcement occurs just before quarter 0, and that β_{t-12} is normalized to be zero. Therefore, for $\tau > -12$, the coefficient $\beta_{t+\tau}$, on the interaction of treatment and quarter $(t + \tau)$, represents the difference between the probability that a treated buyer purchases the recalled product from the treated supplier, and the probability that a control buyer purchases the HS6 product of the control shipment from the supplier of the control shipment in quarter $(t + \tau)$, relative to that difference 12 quarters before the recall announcement.

From $\tau = -12$ to $\tau = -3$, none of the estimated coefficients are statistically different from zero, nor is there any pattern to the point estimates; in other words, the treatment and

control pairs' probabilities of trade follow parallel trends for over two years before diverging. The divergence is abrupt and large: two quarters before the recall announcement, the probability that the treated U.S. firm trades the recalled product with its foreign supplier falls by approximately seven percentage points relative to the control pair's probability of trade. From Table 2, the probability that a given buyer-supplier pair in our sample trades in any pre-recall quarter is about 40%; the 7pp decline therefore represents an economically significant drop of about 17% in the probability of trade. The probability of trade remains 7pp below that of the control in the quarter before the public announcement, then recovers.[15] By the seventh quarter following the recall announcement, point estimates are very close to zero, suggesting that treated and control relationships' survival probabilities converge. However, standard errors are too large to preclude a permanent negative (or positive) long-run difference in the probability of trade of up to about 4pp.

The most compelling explanation for the decline in the probability of trade prior to the public announcement is that the U.S. buyer has private information about the product defect and impending recall.[16] This interpretation is consistent with the process by which CPSC notifies firms when it initiates an investigation into suspected defects or hazards. In this case, we can interpret the subsequent increase in the probability of trade between the affected parties to represent a restoration of trust, possibly following a change in product design, communication or quality control, or suppliers' parts subcontractors.

In Appendix B.2, we explore the robustness of this result to a set of alternative control groups: we add controls that we exclude here because they may be indirectly affected by the recall (Figures B-1 and B-2); we select an alternative set of controls using the supplier

[15] One interpretation of these coefficients is that approximately one in six trade relationships is disrupted for two quarters. Alternatively, it may be that one in three relationships is disrupted for one quarter: half in the quarter just before the recall and half in the prior quarter. We cannot distinguish these two interpretations empirically. Almost all our specification checks in Appendix B show a similar two-quarter pattern.

[16] An alternative, which we cannot distinguish empirically, is that the U.S. buyer is responding to the expectation of the publication of the recall, and not to the underlying cause.

sample (Figure B-3); and we use the U.S. buyer's own prior trading relationships to create a third control group (Figure B-4).[17] Our qualitative, and in many cases quantitative, results hold for these alternative controls; our results are not driven by a particular set of controls or even by a particular set of treatment observations. Across the different control groups, the probability of the treated firms' trade in the recalled product falls by several percentage points in the quarters immediately before the recall announcement. The standard errors on the post-recall announcement coefficients are too large to preclude a permanent negative long-run difference in trading probability.

Three features in Figure 4 and the robustness checks support our identification assumptions. First, the assumption that recalls are plausibly exogenous is supported by the absence of any trend in the first nine quarters in Figure 4. Second, the estimates confirm that the information shock precedes the public recall-announcement date. Finally, the pattern of coefficients suggests that we have successfully identified recalled shipments in the LFTTD.

We next examine whether the lower likelihood of trade with the recalled shipment's supplier is specific to the recalled product. Figure 5 shows the estimated coefficient vector $\widehat{\boldsymbol{\beta}}$, along with 95% confidence intervals, using a different dependent variable: $\mathbb{I}\left(\text{Imports}_{bs,-p,t} > 0\right)$, an indicator that equals 1 iff buyer b trades any *other* product with supplier s in quarter t. Although the pattern of coefficients is somewhat muted, we see a similar dip in the probability of trade in the two or three quarters just before the recall announcement. Different from the recalled product, the probability that the buyer and supplier trade in other products appears to decline starting in quarter (-8) for the treated firms relative to the control firms, although the coefficients prior to $\tau = -3$ are not individually statistically different from zero. That firms appear to reevaluate their entire relationship, at least in the quarters right

[17] Throughout the paper we treat recalls as binary events, although there is considerable heterogeneity in defect severity. In practice, buyers may be less likely to continue relationships where product defects resulted in fatalities than ones that caused only minor injuries. In Appendix B.3.2 and Figure B-6 we test whether recall severity, as proxied by reported injuries in the recall announcement, induces differential responses. We do not find a clear difference between recalls whose announcements included information on injuries or fatalities and those that did not.

around the announcement, suggests that they believe the problem is not simply a random bad batch, but something more systemic, such as quality control at the producing plant. This result is consistent with the notion that recalls convey information about supplier quality. An alternative interpretation is that buyers face economies of scope in purchasing multiple goods from a single supplier, so when they stop purchasing one product due to a recall, they may no longer view that supplier as optimal for other products.[18]

Subsequent to the recall, estimates are less precise, but all post-recall coefficients in Figure 5 are negative and at least half are statistically different from zero, suggesting a likely long-run decline in trade in other products for the treated relationships relative to control relationships. Together with the estimates in Figure 4, this result points to an overall decline in the buyer's reliance on the supplier of a recalled product.

4.3 Spillover Effects on Other Suppliers

Figure 6 shows the coefficients from a regression with dependent variable $\mathbb{I}(\text{Imports}_{b,-s,pt})$, an indicator for the buyer trading product p with a supplier *other than* s in quarter t. From Table 2, more than 60% of buyers in our sample buy the recalled HS6 product from another supplier in any given (pre-recall) quarter: more than half from other suppliers in the same country and about a third from suppliers in other countries.[19]

Two patterns stand out. First, there is a (small and statistically insignificant) increase in treated buyers' probability of trade with other suppliers in the quarters just before the recall announcement, just when the probability of trade with supplier s declines. Second, starting six quarters after the recall announcement, there is a pronounced and statistically significant divergence in the probability that treated and control suppliers trade product p with other

[18] We also cannot rule out the possibility that the recall event has a direct negative impact on some suppliers' production, not only of the recalled product but also of their other products.

[19] These categories are not mutually exclusive. The table also shows that the average buyer in the treatment group has approximately seven suppliers for product p, and the average buyer in the control group has eight.

suppliers. By $\tau = 11$, three years after the recall announcement, treated buyers are ten percentage points more likely to have alternative or additional suppliers for the product.

The next two figures, Figures 7 and 8, separate the effect seen in Figure 6 by the alternative supplier's country of origin. In Figure 7, the probability of trade with another supplier from the same country as supplier s increases by about 6pp by the end of three years; this increase occurs relatively late, two full years after the recall announcement. This finding, that other suppliers in the same country benefit indirectly from a recall, contrasts with Zhao (2018) and Bai, Gazze, and Wang (forthcoming), who find evidence of negative reputational spillovers from product defects to other firms in the same country.

In contrast to same-country trade, the probability of trade with a supplier from a different country, the dependent variable in Figure 8, increases early, possibly starting several quarters before the recall announcement, and reaches an excess 10pp by the end of three years, a 30% increase over baseline levels (from Table 2). This result points to a strong diversification response by the U.S. buyers: they do not merely find other suppliers in the same country but diversify the set of countries with which they trade.

These results and those in Section 4.2 suggest that although buyers can pause trading with little difficulty, it takes time to identify and purchase from alternate suppliers when they discover a problem with an existing supplier. This despite the fact that the types of consumer goods involved in product recalls – clothing, toys, and simple electronics – are less reliant than some specialized intermediate goods on relationship-specific investments and have a very large number of suppliers worldwide.

Taken together, the results also permit us to rule out a possible explanation for the abrupt decline in trade in the six months prior to the recall announcement: that facing a recall, the buyer anticipates a decrease in demand for its products, and curtails orders to avoid having excess inventory. Instead, it appears that the buyer's overall demand is unaffected; only the sourcing of the product changes.

One striking feature of Figure 8 is the positive and statistically significant trend in the

estimated coefficient vector $\widehat{\boldsymbol{\beta}}$ starting as early as three years prior to the recall, which raises the possibility that U.S. buyers may become aware of a problem much earlier than two quarters prior to the recall.[20] If the slow drift towards suppliers in other countries is caused by information about the supplier, it reinforces the long lead time required to identify new suppliers, especially in other countries.

For completeness, we also show, in Figure 9, the effect of a recall on the probability of trade in other products with other suppliers. The difference between treated and control relationships is indistinguishable from zero until the very end of the time period, where, at $\tau = 10$, it increases for treated firms relative to the controls.

The lag of several quarters before U.S. buyers are observed increasing their trade with other suppliers following a recall suggests that searching for new suppliers is costly and time consuming. This interpretation is consistent with evidence that most U.S. trade takes place within established relationships and that, in equilibrium, the cost of establishing new trading relationships is higher than the cost of maintaining existing ones (Monarch and Schmidt-Eisenlohr, 2020; Eaton, Eslava, Jinkins, Krizan, and Tybout, 2014). These higher costs create incentives for firms to invest in intangible, relationship-specific, capital, such as management oversight or communication channels. Although it is not surprising that a recall announcement and the underlying problems it exposes disrupt relationships, we expect longer-lived relationships to have greater trust and communication that can facilitate the resolution of these underlying problems. In the next section, we test the hypothesis that buyers in established relationships are less likely to sever ties with suppliers of recalled products and to form relationships with alternate suppliers.

[20] Mechanically, this coefficient drift could reflect one of two underlying processes. One possibility is that firms do not experience a single information shock, as in Figure 2, but rather a gradual discovery of problems with a supplier. Consequently, the firms are increasingly likely to diversify their supplier portfolio. An alternative, empirically indistinguishable, hypothesis is that the information shock in Figure 2 is an accurate depiction, but the timing of that shock – specifically, the lag between points x and r in the figure – differs across firms in our sample. In the latter interpretation, some firms discover the problem earlier than others, and therefore start reacting sooner.

4.4 New vs. Established Relationships

To explore the hypothesis that relationship age impacts the response to a recall, we estimate the event-study regression from Figure 4 separately for new and established buyer-supplier relationships. We define relationships as new if the buyer-supplier pair has no transactions of the recalled HS6 product in years (-2) and (-3) (i.e., quarters (-12) through (-5)), and as established relationships otherwise. Table A-2 in the Appendix shows that, on average, buyers trade for 6–7 years with their suppliers before a recall announcement.[21]

Estimation results are shown in Figure 10. The coefficient pattern for established relationships, shown as the solid line, mirrors the one in Figure 4: a decline of about 5.5 percentage points in the probability of trade in the two quarters prior to the recall announcement. Also as in Figure 4, the estimated long-run coefficients are near zero but imprecisely measured.

In contrast, new relationships, shown as the dashed line, exhibit larger swings, which bear some explaining. First, every relationship in our sample – new or established, treated or control – must have at least one trade of the HS6 product prior to the recall: for treated relationships, this is the recalled shipment, and for control relationships, this is a matched control shipment. Because new relationships have no trade in the first eight quarters in the graph, they must trade in the four quarters just before the recall. Further, as described in Section 3.2 and Appendix A.2, all control shipments occur within 90 days of the recalled shipment, and the vast majority occur within a much narrower window. Even given these tight match criteria, the graph indicates that treated shipments were 7.5 percentage points more likely than control shipments to occur in quarters $(t-4)$ and $(t-3)$ and 15 percentage points less likely to occur in the two quarters just before the recall. This stark decrease is consistent with the hypothesis that addressing problems in new relationships is more difficult than in existing ones: the drop in trade is much steeper. In the long run, a recall reduces

[21] This statistic is based on the LFTTD for the nine HS2 categories listed in Table 1. The LFTTD begins in 1992, so the variable is truncated.

the probability of trade for new relationships by approximately four percentage points.

Figure 11 shows the dynamics of the probability that the U.S. buyer trades the recalled HS6 product with a different supplier, as in Figure 6, depending on whether the recall affected new or established relationships. In new relationships shocked by recalls, there is a large increase in the probability of the U.S. buyer trading with another supplier in the fourth, third, and second quarters before the recall. Unlike in Figure 10, there is no mechanical reason for this. In the fourth and third quarters prior to the recall, there is no indication that the buyer in a new relationship knows that a recall is impending: in those quarters, the buyer's probability of trade with the supplier of the recalled product is highest. Instead, it appears that these buyers are in some cases experimenting with a new product, which they purchase from multiple suppliers. For established relationships, the likelihood of trading with other suppliers is indistinguishable from zero until almost two years after the recall, when the probability of trading with another supplier increases, consistent with Figure 6. Although the long-run differences between the effects of recalls on new and established relationships are not statistically significant, by three years after the recall, the coefficient estimates for U.S. buyers that experienced a recall in new relationships are twice as large as those that experienced recalls in established relationships: the increase in new relationships is 20pp more than the control vs. 10pp more than the control for established relationships.

4.5 Supplier Reputation

Information about product defects and supplier quality embedded in consumer-product recalls may deter other U.S. buyers from purchasing from the recalled products' foreign suppliers. In this section, we examine potential reputational spillovers on the foreign supplier of the recalled product in the U.S. market. This analysis uses the "supplier sample," which matches treatment and control observations based on the suppliers' characteristics.

We replace the dependent variable in our regressions with an indicator for other U.S. firms purchasing the recalled HS6 product from the recalled shipment's supplier. Figure 12

shows a four percentage-point drop in the probability that the foreign supplier trades with other U.S. buyers one quarter after the recall announcement. The timing is consistent with U.S. buyers gaining additional information about the quality of the foreign supplier from the public announcement. Unlike the buyers of the recalled product, which are contacted by CPSC when the investigation is launched, these buyers do not learn about the recall until it is publicly announced. The fact that the effect is very short-lived could be explained by the supplier taking concrete steps to improve quality control or communication.

In addition, there is suggestive evidence that about seven quarters after the recall announcement, the probability that the foreign supplier trades with other U.S. buyers declines by about five percentage points compared to control suppliers and does not show signs of recovery. From Table 2, the baseline probability of the foreign supplier trading the HS6 product with another U.S. buyer is about 50%, so this decrease amounts to approximately a 10% drop. Overall, evidence of the supplier's other trade relationships responding at all to the recall suggests that the recall imposes some reputational damage on the supplier.

5 Concluding Remarks

In the absence of a Walrasian auctioneer, relationships between buyers and suppliers are the fundamental building blocks of trade. Firms make relationship-specific investments and learn about one another over time. In this paper, we study the repercussions of a negative information shock, whereby a buyer learns about a defect in a product it had purchased from a foreign supplier. Product recalls, caused by the discovery of a defect in production, represent unanticipated shocks to individual trading relationships.

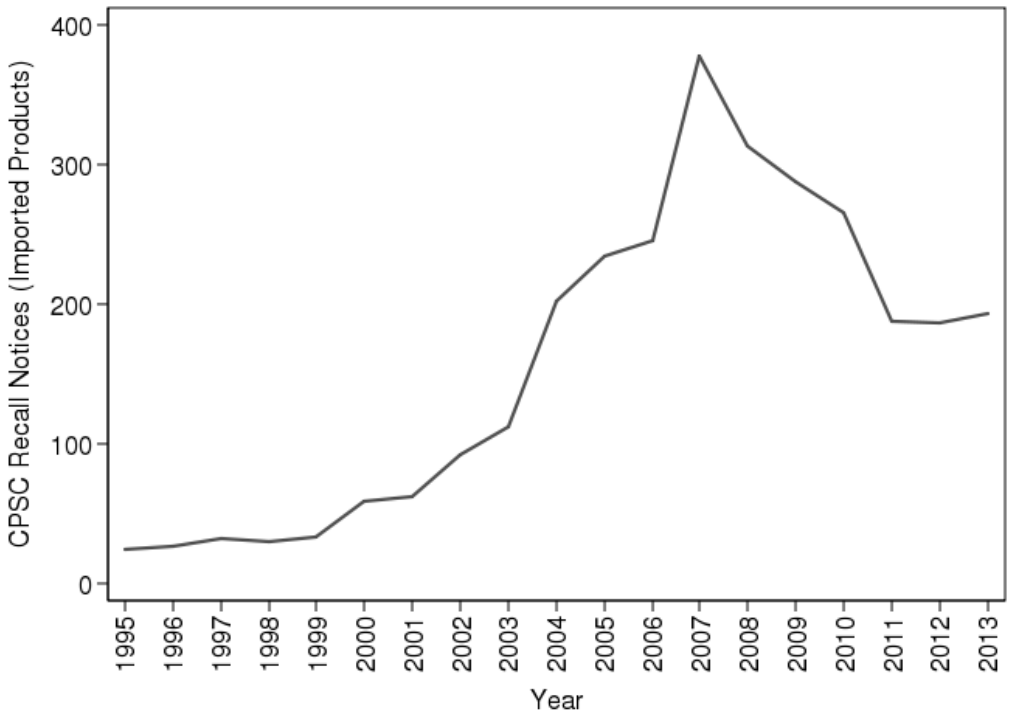
We construct a novel dataset of the universe of U.S. consumer-product recalls sourced internationally between 1995 and 2013, linked to confidential customs transactions records. Using the newly constructed linked data, we find that compared to a control group, treated relationships experience a steep decrease in the probability of trade in the short run. Al-

though the recall affects only one product, trade in other products suffers as well, consistent with the buyer acquiring new information about the quality of the supplier or with economies of scope in purchasing. In the long run, the U.S. buyer increases its reliance on alternative suppliers of the same product, but this shift takes up to two years. We interpret these results to imply that new relationships take time and effort to build. That this is true for relatively simple, mass-produced, consumer goods such as toys and apparel, which have thousands of suppliers globally, implies even higher value of relationships for specialized inputs.

Our results also show how information about a supplier transmits through global supply networks. Most of our evidence points to the U.S. buyer of a recalled product learning about the impending recall when CPSC launches an investigation; some results, particularly purchases from other countries, hint at the possibility that the U.S. buyer may learn about a problem in production even earlier. Other U.S. buyers, not involved in the recall but contracting with the same supplier, only learn about the problem from the public recall announcement. The recall announcement thus not only serves its intended purpose of informing consumers of product defects but also conveys information about the foreign supplier to other U.S. buyers.

The lag of up to two years in U.S. buyers' supplier diversification has important implications for trade policy: firms' responses to changing trade costs may be hindered by their inability to make rapid adjustments to supplier portfolios. For example, sharp increases in U.S. import tariffs in 2018–2019 adversely impacted not only imports but also domestic output and exports of U.S. firms reliant on global supply chains (Flaaen and Pierce, 2019; Handley, Kamal, and Monarch, 2020). Our results suggest that these negative effects may be driven by stickiness in buyer-supplier adjustments.

Figure 1. Recall Announcements of Imported Products, by Year



Source: Authors' calculations from CPSC data.

Figure 2. Stylized Recall Timeline

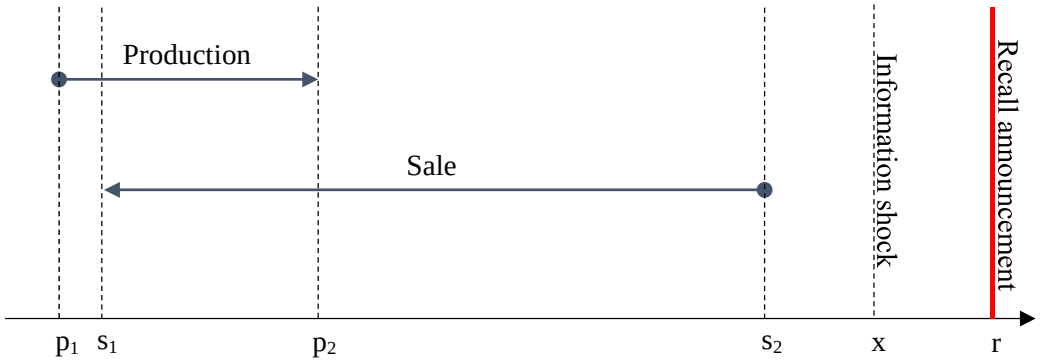


Figure 3. Sample Recall Announcement

Name of product: Toddler sweater fleece pullovers
Hazard: The snaps on the fleece pullovers can detach, posing a choking hazard.
Remedy: Refund
Recall date: September 7, 2017
Recall number: 17-766

Description: This recall involves L.L. Bean toddler sweater fleece pullovers sold in sizes 2T, 3T and 4T. The pullovers have blue trim and were sold in three colors; deep garnet (red), heather gray and light lilac. They have four snaps, two pockets and the L.L. Bean logo tag on the front. Item number 300206 and lot 02/17 are printed on woven tags in attached to the neckline of the pullovers.

Remedy: Consumers should immediately stop using the fleece pullover and contact L.L. Bean for a full refund. L.L. Bean is contacting its customers directly.

Incidents/Injuries: None reported

Sold Exclusively At: L.L. Bean stores nationwide, L.L. Bean catalog and online at www.llbean.com from April 2017 through August 2017 for about $50.

Manufacturer(s): Century Miracle Apparel Manufacturing Co. Ltd., of Jordan

Manufactured In: Jordan

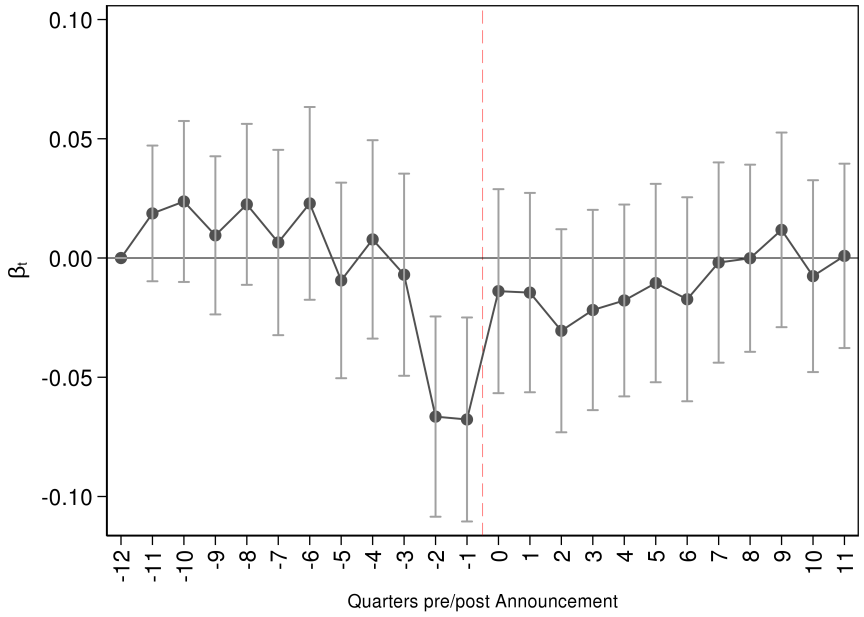
Retailer: L.L. Bean, of Freeport, Maine

Units: About 2,000



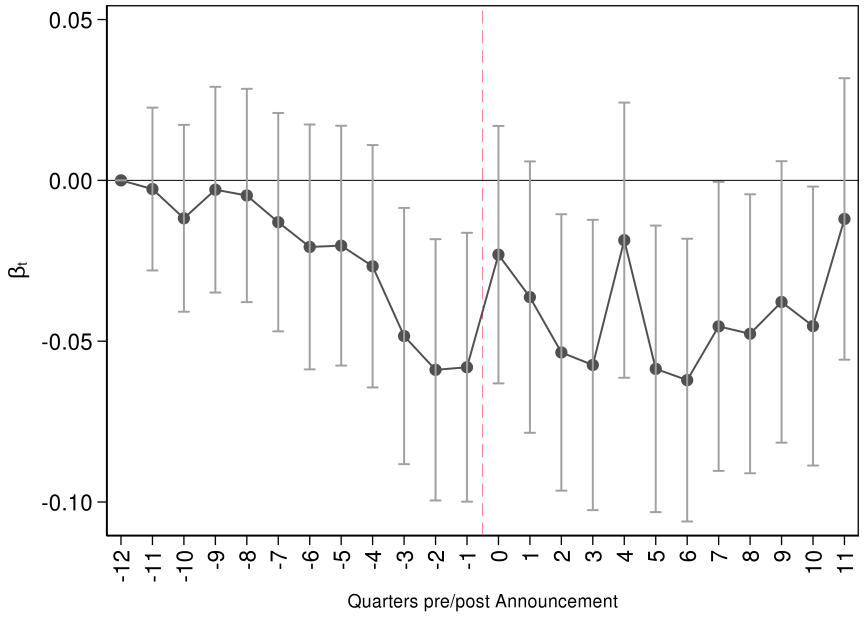
Source: `https://www.cpsc.gov/Recalls/2017/L-L-Bean-Recalls-Toddler-Sweater-Fleece-Pullovers-Recall-Alert`
Note: This recall occurred after the end of our sample period.

Figure 4. Extensive-Margin Dynamics: Buyer-Supplier-Product



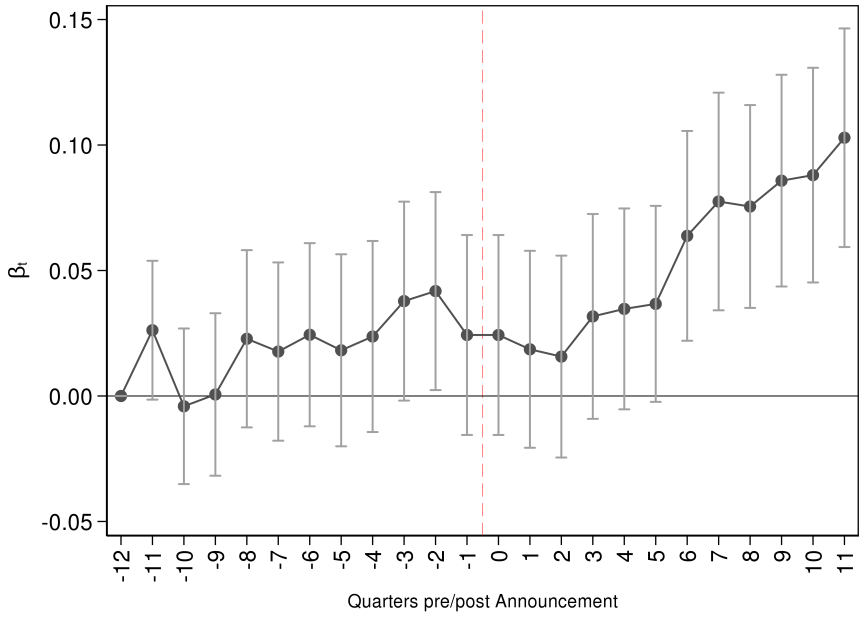
Notes: Coefficient estimates and 95% confidence intervals showing the probability of trade of treated pairs in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control pairs. The regression includes quarter and buyer-supplier fixed effects. Standard errors are clustered by recall.

Figure 5. Extensive-Margin Dynamics:
Buyer-Supplier-Other Product



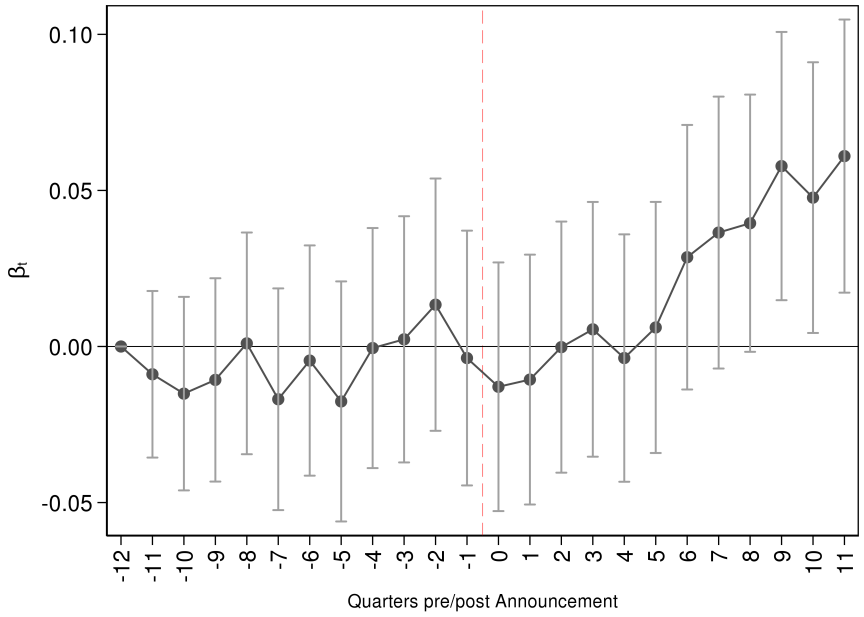
Notes: Coefficient estimates and 95% confidence intervals showing the probability of trade between of treated pairs in all non-recalled HS6 products, for 12 quarters before and after a public recall announcement, relative to control pairs. The regression includes quarter and buyer-supplier fixed effects. Standard errors are clustered by recall.

Figure 6. Extensive-Margin Dynamics: Buyer-Other Supplier-Product



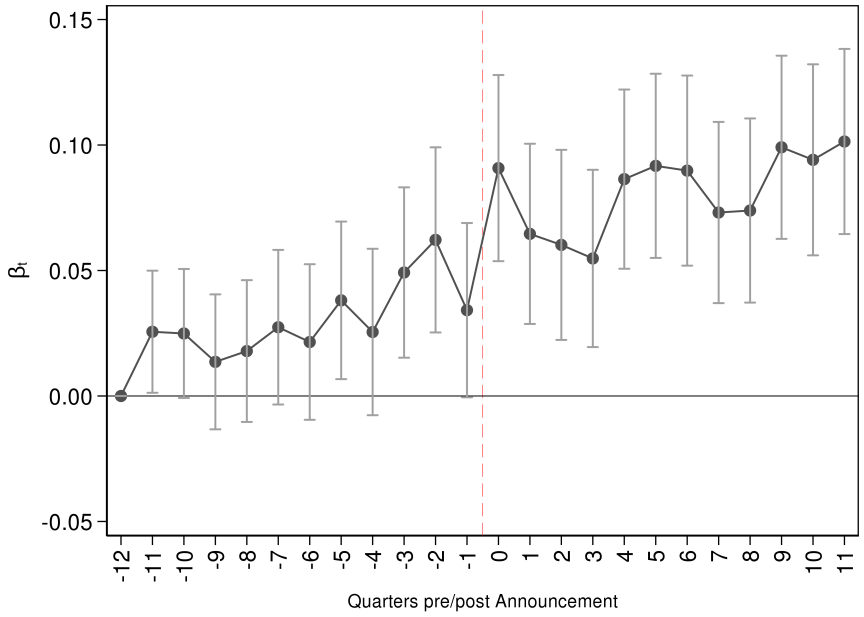
Notes: Coefficient estimates and 95% confidence intervals showing the differential probability of trade between the treated buyer and other suppliers in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control pairs. The regression includes quarter and buyer-supplier fixed effects. Standard errors are clustered by recall.

Figure 7. Extensive-Margin Dynamics: Buyer-Other Supplier, Same Country-Product



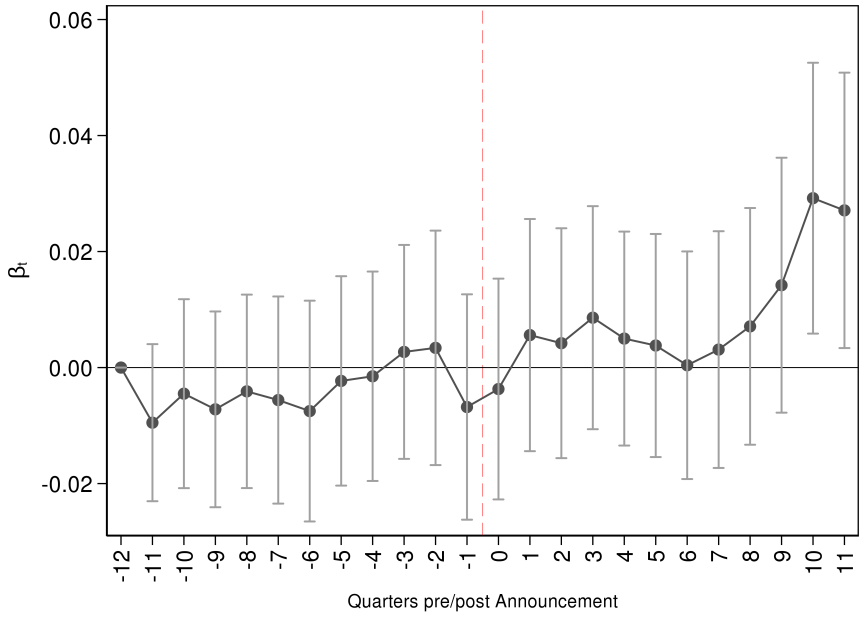
Notes: Coefficient estimates and 95% confidence intervals showing the differential probability of trade between the treated buyer and other suppliers from the same country in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control pairs. The regression includes quarter and buyer-supplier fixed effects. Standard errors are clustered by recall.

Figure 8. Extensive-Margin Dynamics: Buyer-Other Supplier, Other Country-Product



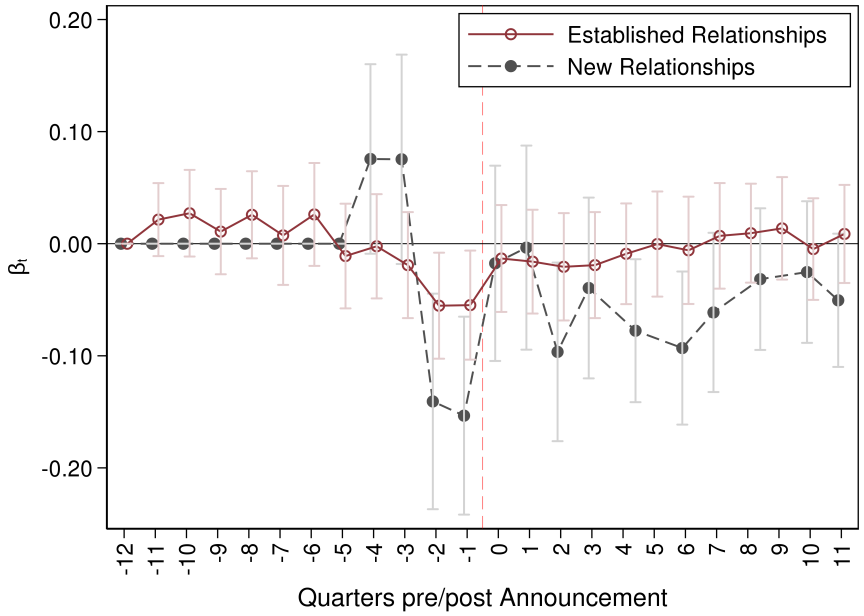
Notes: Coefficient estimates and 95% confidence intervals showing the differential probability of trade between the treated buyer and suppliers from other countries in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control firms. The regression includes quarter and buyer-supplier fixed effects. Standard errors are clustered by recall.

Figure 9. Extensive-Margin Dynamics: Buyer-Other Supplier-Other Product



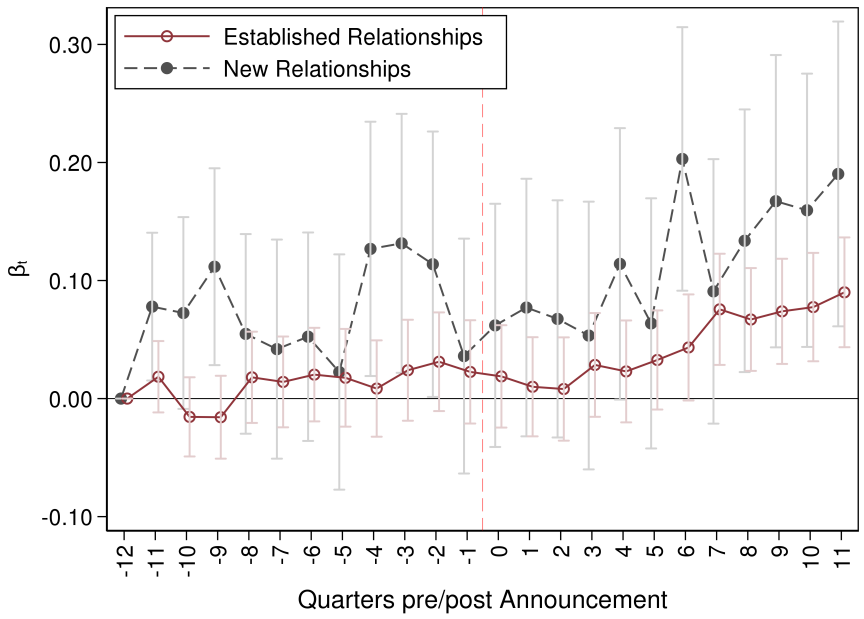
Notes: Coefficient estimates and 95% confidence intervals showing the differential probability of trade between the treated buyer and other suppliers in all non-recalled HS6 product categories, for 12 quarters before and after a public recall announcement, relative to control pairs. The regression includes quarter and buyer-supplier fixed effects. Standard errors are clustered by recall.

Figure 10. Extensive-Margin Dynamics: Buyer-Supplier-Product, for Established vs. New Relationships



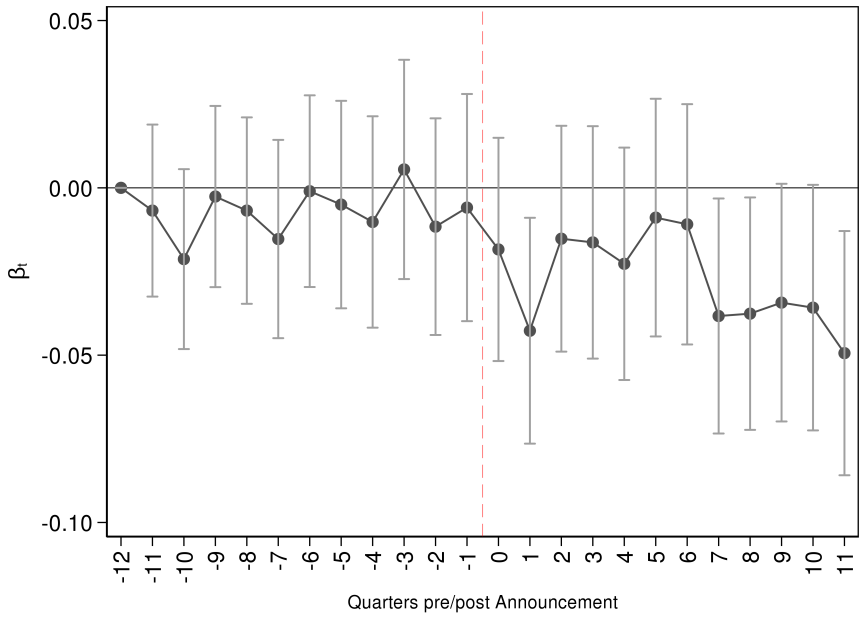
Notes: Coefficient estimates and 95% confidence intervals showing the probability of trade of treated pairs in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control pairs. New suppliers are those with which the firm has zero trade in quarters (-12) through (-5); established suppliers are all other supplies. The regression includes buyer-supplier and quarter-by-relationship-type fixed effects. Due to small cell sizes, some coefficients have been combined. Standard errors are clustered by recall.

Figure 11. Extensive-Margin Dynamics:
Buyer-Other Supplier-Product, for Recalls Affecting
Established vs. New Relationships



Notes: Coefficient estimates and 95% confidence intervals showing the probability of trade between the treated buyer and other suppliers in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control firms. New suppliers are those with which the firm has zero trade in quarters (-12) through (-5); established suppliers are all other supplies. The regression includes buyer-supplier and quarter-by-relationship-type fixed effects. Standard errors are clustered by recall.

Figure 12. Extensive-Margin Dynamics: Supplier-Other Buyer-Product



Notes: Coefficient estimates and 95% confidence intervals showing the differential probability of trade between the treated foreign supplier and all other U.S. buyers not impacted by a recall in all recalled HS6 product categories, for 12 quarters before and after a public recall announcement, relative to control pairs. The regression includes quarter and buyer-supplier fixed effects. Standard errors are clustered by recall.

Table 1. Recalled Shipments, by HS2 Product Classification

HS2	HS Description	Recalls	$\frac{\text{Recalls}}{\text{Shipments}}$ (%)	$\frac{\text{Matched Recalls}}{\text{All Recalls}}$ (%)
39	Plastic and articles thereof	171	0.000874	28.07
61	Apparel, knitted or crocheted	258	0.001461	36.43
62	Apparel, not knitted or crocheted	183	0.000833	29.51
73	Articles of iron and steel	122	0.001101	32.25
94	Machinery and mechanical appliances	250	0.000594	40.40
85	Electrical machinery and equipment parts	557	0.001178	46.50
87	Vehicles other than railway or tramway	290	0.002118	38.28
94	Furniture, bedding, mattresses, mattress support, cushions, lamps and lighting fittings	395	0.002207	47.09
95	Toys, games, and sports requisites	947	0.011100	39.60

Notes: Statistics for the HS2 products representing three-quarters of all imported consumer-product recalls. The first column shows the number of imported consumer-product recalls (from CPSC) by each HS2. The second column shows recalls as a percentage of total imported shipments between 1995 and 2013 (from LFTTD). An import shipment is defined as a combination of a U.S. buyer-foreign supplier-HS10-date. A single recall may map to more than one HS2; such recalls are double counted in this table. The final column shows imported consumer-product recalls linked to the LFTTD as a percentage of total imported consumer-product recalls for each HS2.
Source: Authors' calculations from CPSC and LFTTD data.

Table 2. Summary Statistics

	Treatment	Control
A. Buyer Sample[a]		
$\mathbb{I}$(Imports HS6 product from supplier)	0.4074	0.3839
	(0.3702)	(0.3548)
$\mathbb{I}$(Imports other products from supplier)	0.5154	0.5146
	(0.4211)	(0.4060)
$\mathbb{I}$(Imports HS6 product from other suppliers)	0.6248	0.6908
	(0.3968)	(0.3806)
$\mathbb{I}$(Imports HS6 product from other suppliers, same country)	0.5256	0.6101
	(0.4135)	(0.4067)
$\mathbb{I}$(Imports HS6 product from other countries)	0.3330	0.3742
	(0.4110)	(0.4205)
$\mathbb{I}$(Imports other products from other suppliers)	0.9653	0.9742
	(0.1365)	(0.1072)
Value of buyer's transactions with supplier[b]	3,247	7,004
	(54,670)	(173,100)
	[51.14]	[44.96]
Value of buyer's transactions with other suppliers[b]	100,500	60,270
	(384,500)	(179,700)
	[6,252]	[5,714]
B. Supplier Sample		
$\mathbb{I}$(Exports HS6 product to buyer)	0.4057	0.3867
	(0.3630)	(0.3575)
$\mathbb{I}$(Exports HS6 product to other buyers)	0.5020	0.4695
	(0.4271)	(0.4210)

Notes: Mean (with standard deviation in parentheses and medians in square backets) in quarters $\{-12, -9, -10, -11, -8, -7, -6, -5\}$ relative to the recall announcement, for treatment and control groups. Medians are approximate: they show the average of the values for firms between the 45th and 55th percentiles of the variable's values. $\mathbb{I}(\cdot)$ is an indicator variable. All numbers have been rounded to comply with Census Bureau disclosure-avoidance rules.

[a] Based on 750 treated relationships and 4,200 control relationships per quarter.

[b] Thousands of 2015 U.S. dollars.

A Data Appendix

A.1 Identifying Recalled Shipments

We use the product name and detailed description to attach one or more possible HS10 codes to each recall. We assign multiple possible HS10 codes to recalls when we cannot be certain which code would have been used in the customs forms. For example, apparel items have different HS codes depending on their material (cotton vs. synthetic fibers), and this information is not always available in the recall description. As another example, musical instruments are classified under HS code 9205, but toys are classified under HS code 9503; we cannot always tell whether a "toy drum set" or "toy xylophone" is classified as a toy or as a musical instrument. As a final example, some recalls are of bundled products, such as candles and candle-holders, which may have been assigned separate HS codes at entry.[22] More than half of the HS codes we assign to recalls fall into four two-digit HS codes: HS 95 (toys, games, and sports equipment); HS 85 (electrical machinery and equipment); HS 61 (articles of apparel and clothing, knitted or crocheted); and HS 94 (miscellaneous manufactured articles; primarily miscellaneous furniture items and mattresses).

Since the recall notices do not contain the address information of the foreign supplier, we create an abbreviated supplier firm name by concatenating the first three letters of the first word with the first three letters of the second word in the name. Following the instructions on the customs form for reporting MIDs (Form-7501), in cases where the supplier has a one-word name or where the second word of a supplier's name is fairly generic, like "Industries" or "Corporation," or simply repeats the name of the country, we use the first three letters of the first word only.

The recall announcement includes information about the range of dates during which

[22] Whenever possible, we rely on a list of HS codes provided by CPSC under the "eFiling Alpha Pilot" to assign HS codes to recalled products. This project is intended to increase CPSC's import targeting capabilities. See `https://www.cpsc.gov/Imports`.

the product was sold (more than 95% of cases) and/or produced (under 5%). Dates ranges can be exact (e.g., "from January 1 to January 31, 1999"), or vague (e.g., "spring of 1999"). In the latter case, we translate them into calendar dates, being as broad as possible.

From the press release associated with the recall depicted in Figure 3, we would extract the following information:

- Announcement date: September 7, 2017
- U.S. buyer: L.L. Bean
- MID: CENMIR
- Country: Jordan
- HS code(s): 6110, 6111[23]
- Sale date(s): April 2017 to August 2017
- Production date(s): n/a

Table A-1 reports the unique number of recalls (in the first column) and the unique number of U.S. importing firms (in the second column) that we link at each step of our matching process. A single recall may be linked to multiple firms, and a single firm may be linked to multiple recalls. Our goal is to identify the shipment in the LFTTD associated with a recall at the buyer-supplier-country-product-date level.

We begin by searching the BR for firms with names identical or similar to the U.S. buyers listed in the recall announcements in step (1). We match 90% of the names associated with a recall to the BR.[24] For the subset of firms identified as possible matches, we search the LFTTD to determine whether they have ever imported any products with one or more of the identified HS codes from the country from which the recalled product was imported in step (2). In step (3), where possible, we restrict to shipments that ever transacted with

[23]HS 6110 is "Sweaters, pullovers, sweatshirts, waistcoats (vests) and similar articles, knitted or crocheted;" 6111 is "Babies' garments and clothing accessories, knitted or crocheted." As this is a toddler's vest, we cannot be certain whether the adult or baby HS code was used.

[24]This figure compares favorably with other studies merging external data to the BR; see Basker and Simcoe (2021, Appendix A.1) for a discussion.

a firm with an abbreviated supplier name similar or identical to the abbreviated supplier name identified in the announcement. Supplier names do not always appear in the recall announcements. If none of the shipments from step (2) match on the supplier name, we do no culling at this point. Consequently, step (3) drops no recalls but does eliminate about half of the remaining candidate firm matches.

In step (4), we restrict shipments using date information. Because some products are imported in one shipment and others are imported over years, in recurring shipments, our goal is to find the last possible shipment for any buyer-supplier-product tuple. We start by dropping shipments that occurred after the recall-announcement date. We also limit our attention to shipments that occurred on or after the first production date, if provided in the recall announcement, and no earlier than 365 days prior to the first sale date if no production date is available. We further limit ourselves to shipments that occurred no later than the last sale date, which is usually provided in the announcement. In rare cases the recall announcement includes information about import dates; in those cases, we limit attention to shipments in that window. For illustration purposes, consider again the stylized case depicted in Figure 2. The announcement takes place at time r, so we know the recalled shipment had to have occurred before r. The product was sold during the interval $[s_1, s_2]$, so the last shipment had to have occurred before s_2. If the recall announcement includes a range $[p_1, p_2]$ for production, we can also infer that the recalled shipment had to have occurred after p_1.

Even after applying these restrictions, some recall announcements match to many different possible buyer-supplier pairs, whereas others match to just one (or in some cases, none). We limit our analysis to recalls where we identify the buyer and supplier with no more than ten possible buyer-supplier-country-HS6 matches, as shown in step (5). In some cases, we may identify a single buyer-supplier pair, but several candidate HS6s; in other cases, we may identify one or two buyers and a single HS6, but several possible suppliers.

We impose two additional restrictions to create our analysis sample. First, we limit the

sample to cases where the buyer has at least one trade (of any product, with any supplier) in each quarter over the four years surrounding the recall announcement, as shown in step (6). Our goal with this restriction is to remove firms that have very little trade, including firms that have already gone out of business before the recall was issued, from the analysis, since they cannot help with identification of a recall effect.[25] Second, we only retain shipments for which we were able to find at least one control shipment, as shown in step (7). (The process of identifying control shipments is described in Appendix A.2.) The last two rows in Table A-1 split our matches into those that are associated with a single and multiple shipments in the LFTTD, respectively. The bulk of our analysis sample contain recalls that were matched to multiple possible shipments in the LFTTD.

A.2 Identifying Control Shipments

We implement the following two-stage procedure to find plausible controls for our analysis sample: first, we identify possible controls at the shipment level; second, we restrict the set of controls based on shared firm characteristics.

We start by identifying one or more possible **control shipments** for each recalled shipment. Control shipments must meet the following criteria: they were shipped within a 90-day window on either side of the recalled shipment, but before the recall announcement date; the country of origin is the same as for the recalled shipment; the product is the same at the HS4 level but has a different HS6. We drop any control shipments whose buyer or supplier directly experienced any recall, as well as any control shipment whose foreign supplier ever traded with the buyer of the recalled shipment in the matched treatment observation or whose U.S. buyer ever traded with the supplier of the matched recalled shipment.[26]

These control shipments have the advantage that the control buyers and suppliers are not directly impacted by the recall, and that idiosyncratic effects associated with unobservable

[25] We relax this condition in Figure B-2.

[26] We relax these restrictions in a robustness test in Appendix B.2.

country, HS4, and time-period characteristics are identical in the treatment and control shipments. In particular, the control shipments share the treated shipments' industry, global and country-specific conditions, and the country's trade and exchange-rate policies.

Restricting the controls so that they do not share an HS6 or ever trade with the recalled shipment's supplier or buyer is intended to mitigate the concern that controls may be *indirectly* impacted by recalls. A recall concerns a product supplied by supplier s to buyer b; the control relationships involve two completely different firms, supplier s' and buyer b'. In addition, control shipments and treated shipments trade the same HS4, but this is a fairly broad category.[27]

Because the U.S. buyers in the treated and control samples are different, there is a risk that differences in outcomes are driven by differences in buyer-level characteristics. To address this concern, we restrict buyers in the control set to match the buyer of the treatment shipment on several characteristics: (i) affiliation status (exact match); (ii) total value of the buyer's imports in the three years prior to the recall announcement (window match); (iii) value of buyer's imports from the supplier in the three years prior to the recall announcement (window match); (iv) supplier's share of the buyer's total imports in the three years prior to the recall announcement (window match); (v) product share of buyer-supplier trade in the three years prior to the recall announcement (window match).

For affiliation status, we match recalls involving affiliated buyer-supplier pairs to control shipments by other affiliated buyer-supplier pairs, and recalls involving unaffiliated buyer-supplier pairs to control shipments by other unaffiliated buyer-supplier pairs.

To match on total value of trade and value of buyer-supplier trade, we bin trade values into eight bins: less than \$10,000, between \$10,000 and \$100,000, and so on, with the largest bin capturing firms with more than \$10 billion in trade. To match on the shares (supplier

[27] An example of an HS4 code is 6106, "Women's or girls' blouses and shirts, knitted or crocheted." Within this HS4 code, HS 6106.20 is "[...] of man-made fibers," and HS10 code 6106.20.10.10 is the most specific: "Women's blouses and shirts, knitted or crocheted, of man-made fibers, containing 23% or more by weight of wool or fine animal hair."

as a share of buyer's total imports, and product as a share of the buyer-supplier trade), we generally match within 20 percentage points. The exception is that, when the share (of the relationship relative to the buyer's total imports, and of the product relative to the relationship's total trade) is zero, we match to a control with a zero share as well. In addition, if we retain more than ten control shipments for any treated shipment after making the above restrictions, we apply a 10pp cutoff instead of the 20pp cutoff.

Finally, if we have more than ten control shipments for a given treated shipment after applying all the above restrictions, we restrict the control shipments to the ten closest in date to the recalled shipment's date. As a result, the vast majority of control shipments occur within a month of the recalled shipment.

Table A-2 shows additional summary statistics for the treatment and control observations. The treated supplier accounts for about 12% of the treated buyer's overall imports in the average quarter; the corresponding figure for control relationships is 10%. The supplier accounts for a larger share of the HS6 product's imports: 32% for treated relationships, 25% for controls. As a share of the pair's total value of trade, the HS6 product accounts for 40% in treated, and 36% in control, relationships. The average number of suppliers with which the U.S. buyer trades per quarter is about 340 for treated and 450 for control buyers. Treated buyers purchase the HS6 product from seven suppliers in any given pre-recall quarter; control buyers have eight suppliers. The U.S. buyers in both the treated and control groups import more than 90 unique HS6 products, from 11–12 source countries, per quarter. A treated buyer-supplier pair in our sample transact for seven years prior to experiencing the first recall. For control pairs, the equivalent is six years. For comparison, the average buyer-supplier relationship in the full LFTTD transacts for about four.[28]

The bottom panel of Table A-2 shows the number of HS6 products that are sold by a foreign supplier in the supplier sample. Treated suppliers sell 50 and control suppliers sell

[28]This number is based on the nine HS2 categories listed in Table 1.

55 HS6 products to the U.S. in an average quarter.

A.3 Manufacturer Identifiers

MIDs may be subject to clerical errors. Importantly, we assume that these errors are unrelated to recall events. Errors may occur due to variations in the name and address information reported on customs forms, for example, if a single supplier's MID is constructed sometimes using its full name (e.g., Basker-Kamal Enterprises) and elsewhere using an abbreviated name (e.g., BKE), or if the same foreign name is transliterated in multiple ways. Such MID variations may cause attenuation bias in some of our analyses. If a single buyer always uses the same MID to refer to a single supplier, the buyer-supplier level regressions may not be impacted, but our estimates of the impact on other buyers of the same supplier may be attenuated. Relatedly, constructing abbreviated supplier identifiers implies that we bundle a large number of potentially unique MIDs in the LFTTD.

It is impossible to be certain how much of the proliferation of MIDs is due to typos and errors, multiple establishments within a firm, legitimately distinct firms, or changes in a firm name over time. Our modified MID is intended to capture the concept of a firm rather than an establishment (DeSalvo, Limehouse, and Klimek, 2016); our aim in aggregating is to remove as much as possible the typos and errors, as well as establishment-specific identifiers, to focus on the firm. (In the CPSC dataset there is no way for us to separately identify an establishment for matching purposes.) On average, there are approximately 3.2 MIDs per abbreviated supplier code, of which the largest accounts, on average, for 88% of the trade associated with these MIDs over the full period 1992-2015.[29]

We believe that our aggregation choice is appropriate based on the *ex-post* sharp differences between the same-supplier (Figure 4) and the other-supplier, same-country (Figure 7)

[29]Among abbreviated supplier codes associated with more than one MID, there are approximately 6.5 MIDs per abbreviated supplier code, of which the largest MID accounts for 70% of trade. As a point of comparison, published numbers from the 1997 U.S. Economic Census show that the average manufacturing firm with more than one establishment had 3.2 establishments (U.S. Census Bureau, 2001, Table 3-3).

results: we find a 7pp short-run decline in same-supplier probability of trade in the short run and an ambiguous long-run effect; and a 6pp long-run increase, but no short-run effect in the probability that the U.S. buyer trades with other suppliers from the same country.

B Additional Results

B.1 Predicting Recalls

Are recalls predictable? Specifically, does having one recall signal a firm's increased propensity to experience subsequent recalls?

To address this question, we use the 2007 LFTTD and CPSC data. We chose 2007 because that is the year with the highest number of recalls in our data; even then, fewer than 0.01% of shipments were recalled. We limit our analysis to shipments in the nine HS2 codes with the highest number of recalls, and aggregate individual shipments to the buyer-supplier-HS10-date level.[30] We estimate

$$\text{Recall}_{bspt} = \alpha_p + \delta_{c(s)} + \beta \text{PriorRecalls}_b + \gamma_1 \ln(\text{PriorValue}_b) + \gamma_2 \mathbb{I}(\text{PriorValue}_b > 0) + \varepsilon_{bspt}$$

where Recall is the probability that a shipment from supplier s to buyer b of product (HS10) p at time t is recalled (described in more detail in the next paragraph); α_p is a product fixed effect; $\delta_{c(s)}$ is a fixed effect for the supplier's country of origin; PriorRecalls is the number of buyer b's prior recalls from 1992 to 2006; and PriorValue is the total value of imports by buyer b (1992–2006, deflated using the monthly all-products CPI). The error term is clustered at the HS10 level. The fixed effects are intended to capture the possibility that some products or countries are more prone to defects and recalls than others.

[30] The HS2 codes are listed in Table 1. The aggregation combines shipments that arrive at the same or different ports on the same day, from the same source firm to the same destination firm, with a single HS10 code. We cannot do this analysis with the full LFTTD for computational reasons. In the 2007 file alone, even after our restriction on HS2 codes, we have more than 13 million observations.

The variable Recall, capturing the probability that a shipment is recalled, is constructed as follows. If we link a single CPSC announcement to a single shipment, this variable has a value of 1. If we identify n shipments as potential match candidates – for example, shipments representing multiple HS10 codes within a single HS6, or shipments representing several different buyers or suppliers – the variable Recalls take the value of $1/n$ for each of the n shipments. Finally, because recalls are so rare, we multiply the variable by 100,000.

Our interest is in the coefficient β on the number of prior recalls. Because there may be a mechanical relationship between the number of prior recalls and the total value of prior shipments, we also include the log of the prior value of all shipments. For firms with no imports in 1992–2006, we arbitrarily set the log of the value of total shipments to zero, and add an indicator to capture the difference between firms with no prior shipments and firms with prior shipments valued at \$1.[31]

We do not interpret the coefficients causally: prior recalls are not the cause of current recalls. Rather, our interest is in examining the degree of persistence in the propensity of firms to experience recalls. A positive coefficient β indicates that firms that previously experienced a recall are more likely to experience another one, perhaps because of poor quality-control protocols, because they import products particularly susceptible to defects (even within the HS10 product category), or because they arise suspicion and are more likely to be inspected by CBP or CPSC.

The results are shown in the first column of Table B-1. The probability that a buyer's shipment is recalled is not correlated with the probability that a prior shipment was recalled: the coefficient of 0.001 implies that each additional prior shipment that the buyer had recalled is associated with an increase of $1/100{,}000{,}000$ in the probability that a current shipment is recalled.

In column (2) we replace PriorRecalls_b and PriorValue_b by PriorRecalls_s and PriorValue_s:

[31] As noted in Section 3.1.1, the LFTTD includes only imported shipments valued over US\$2,000.

the probability that a *supplier's* prior shipment was recalled, and the supplier's total value of prior shipments. Here the coefficient is considerably larger, but still very small in absolute value and not statistically significant. The mean probability that a shipment is recalled, from Table 1, is approximately 0.002%; the coefficient of 0.095 implies that a supplier's prior recall is associated with an increase in the probability of a current recall of less than 0.0001%, or one hundredth of the mean.

In column (3) we replace PriorRecalls_b and PriorValue_b by PriorRecalls_{bs} and PriorValue_{bs}: the probability that the buyer-supplier pair had a prior recall, and the pair's prior trade value. The coefficient β is ten times larger than in the prior column, but not statistically significant.

Finally, in column (4) we include all three prior recall counts – for the buyer, the supplier, and the buyer-supplier pair – as well as all the prior trade values. All three coefficients are larger, but none are statistically significant.

These regressions are all conditional on a future shipment to a U.S. buyer: if the supplier's reputation is sufficiently sullied, because the defect that precipitated the original recall is deemed to be a signal of its poor quality, its orders may dry up and we will not observe additional shipments in our data. In that sense the selection effect may under-estimate the true extent to which a supplier's prior recalls reflect upon its future quality.

These results demonstrate that recalls are largely unpredictable, at least within a narrow product and country classification. Our interpretation of this finding is that treating recalls as quasi-exogenous in the main specifications in the paper is reasonable. Nevertheless, all of our specifications include buyer-supplier fixed effects to allow for the possibility that some buyers, suppliers, or buyer-supplier pairs have a higher propensity to experience recalls. As long as this propensity is time-invariant, reflecting the underlying quality of the supplier, relationship-specific fixed effects in our event-study regressions in the paper should be sufficient to control for differences between treated and control relationships.

B.2 Alternative Controls

The sample used to create Figure 4 excludes control observations in which (1) the U.S. buyer had ever traded with the treated supplier; (2) the foreign supplier had ever traded with the treated buyer; or (3) the HS6 product in the control shipment is identical to the HS6 product in the treated relationship. We make these restrictions out of an abundance of caution to ensure that the results are not contaminated by indirect effects of recalls on control relationships. In Figure B-1, we add these control observations. Coefficient estimates are qualitatively similar, but the magnitude of the drop in the probability of trade in the two quarters prior to the recall is about one percentage point larger, at 8pp compared to 7pp in Figure 4.[32]

The sample in Figure 4 also restricts both treated and control firms to those that have had at least one trade transaction – with any supplier, for any product – in each of the eight quarters just before and just after the recall announcement. The purpose of dropping firms that are completely inactive from the analysis, in step (6) of our sample creation (Table A-2), is to allow us to focus on the survival of relationships separate from the survival of the firms themselves. In Figure B-2 we add inactive firms to the analysis. This increases the sample size by about a third. Again, the qualitative results are similar, but this time the coefficients are slightly attenuated relative to Figure 4, showing about a 5.5pp drop in the probability of trade in the two quarters before the recall announcement. The direction of change is as expected; we have added firms that have extremely low (zero) probability of trade in any quarter, so the difference between quarters is muted.

Finally, we estimate Equation (1) on two alternative control groups. The first uses the supplier sample. Controls are selected using the same strategy outlined in Section 3.2 in the paper but matched on the characteristics of the foreign suppliers rather than the U.S. buyers. After selecting control shipments, we narrow down the set of control relationships

[32]Due to disclosure-avoidance concerns, we cannot break out the three types of additional controls, but the majority are of the third type.

to match: (i) affiliation status (exact match); (ii) total value of the supplier's exports to the U.S. in the three years prior to the recall announcement (window match); (iii) value of supplier's exports to the buyer in the three years prior to the recall announcement (window match); (iv) buyer's share of the supplier's total exports to the U.S. in the three years prior to the recall announcement (window match); and (v) product share of buyer-supplier trade in the three years prior to the recall announcement (window match).

Results from this regression are shown in Figure B-3. These results appear extremely similar to the ones shown in Figure 4 in the main text.

The second alternative control group takes a different approach entirely. Instead of selecting control shipments traded by different firms around the same time period, we find control shipments purchased by the same U.S. buyers as the treated shipment (but from different suppliers), in the same broad product classification (HS2) two years ($\pm$90 days) prior to the recalled shipment. This control group has important advantages, as well as some disadvantages, over the main control group. First and foremost, by construction, the control relationship – two years prior to the recall – cannot have been impacted, directly or indirectly, by the recall. The fact that it was imported by the same U.S. buyer means that any unobserved, but time-invariant, buyer-level traits, such as the longevity of its relationships, is implicitly controlled for.

The set of buyers for which we find a control shipment meeting these criteria is relatively small, limiting our power. This sample is also selected to skew larger and older than the main sample because a firm has to have similar shipments two years apart to be included. Finally, because the treatment and control observations are from different time periods, they may be affected by different external factors such as macroeconomic conditions, exchange-rate fluctuations, or shocks to industry demand.

To use this alternative control, we must contend with survivor bias. By construction, any firm that is included in this sample must survive (and continue trading similar products) for at least two years after its *control* shipment, because the control shipment is dated two

years before the recalled shipment. To remove this bias, we further restrict the sample to firms that also have at least one shipment two years after the recalled (treated) shipment.

Results for the second alternative control are shown in Figure B-4. These results, although consistent with the main sample, are much less precise – note that the range of the y-axis is twice as large as in Figure 4. There is a larger, but statistically weaker, decrease in the probability of trade with the supplier of the recalled shipment, relative to the probability of trade with the control supplier, one or two quarters prior to the recall announcement. Different from the result in Figure 4, the relationship appears to take a couple of quarters longer to start recovering. Here, the long-run effect is unambiguously negative: although almost none of the coefficients are individually statistically significant, the pattern clearly indicates a long-run reduced trade probability.

B.3 Additional Results

B.3.1 Intensive Margin

To see whether the intensive margin of trade is affected by recalls, we estimated Equation (1) with dependent variable $\ln(\text{Imports}_{bst})$, the log value of imports by buyer b from supplier s in quarter t, capturing the intensive margin of adjustment, combining the recalled product and all other products. This variable is defined only for firms that have nonzero imports from that supplier in that quarter, so the sample is not a balanced panel.[33] Figure B-5 shows the coefficient estimates. A few coefficients are statistically significant, but there is no clear pattern in the value traded between pairs before and after the announcement.

[33] An alternative to log value, increasingly popular, is the inverse hyperbolic sine (IHS) of value. This variable has the advantage that it behaves much like the natural logarithm for nonzero values but is still defined for zero values; in both senses it is similar to $\ln(x+1)$. In cases like ours, in which a large share of the observations are zeros, regression results using IHS of value are dominated by the extensive margin and do not shed additional light on the intensive margin.

B.3.2 Recall Severity

In this section we probe for heterogeneous effects by the nature of a recall. Although it would have been instructive to understand how the cause of a recall (such as negligence in manufacturing or faulty design) impacts the likelihood of trading, the recall notices do not contain this information. As in the example in Figure 3, recall notices do typically include reports of injuries associated with product defects. Such injury reports may signal the severity of a product defect. Defect severity could have differential impacts on the speed of response, the magnitude of the short-run decline in trade probability, or the long-run probability of relationship survival.

In practice, injuries are at best a noisy measure of defect severity. Some notices include reports of injuries or fatalities; other do not, but the defects may have caused unreported injuries.[34] In other cases, serious product defects could have been detected through quality-control checks, or action may have been taken quickly before any injuries occurred. In addition, in some years CPSC recall notices include very few indications of injuries, possibly because its database did not capture that information.

Nevertheless, as reported injuries are the only metric we observe associated with severity, we re-estimate Equation (1) separately for recalls with and without reported injuries or fatalities. The results are shown in Figure B-6. We do not observe either economically or statistically different patterns for the two sets of recalls. For recalls with reported injuries (solid line), point estimates suggest that the decline in the probability of trade begins earlier, possibly as early as a full year before the recall. Although firms are required to report injuries to CPSC within 24 hours, and the recall process is likely to be expedited in such cases, it is possible that recalls that eventually produce injuries have some non-reportable precursors that prompt firms to evaluate their trade relationships. The decrease from quarter (-4) to

[34] Examples of reported injuries include "three injuries with burns requiring hospitalization" (Recall Number 12007) and "Seven injuries to children include damaged teeth, stitches to the chin, cuts and scratches to the mouth and face" (Recall Number 2175). These recalls are of domestically sourced products, and are therefore not in our sample.

quarter (-2) is only marginally significant statistically, so we cannot reject the possibility that it is random.

Table A-1. Matching Recalls to LFTTD

Step		Recalls	Firms
(1)	Match to BR	2,700	45,000
(2)	Share of (1) that match to LFTTD by HS6 + country	73.37%	5.38%
(3)	Share of (2) that also match by MID	100.00%	50.87%
(4)	Share of (3) that also match by date	65.55%	67.45%
(5)	Share of (4) that match no more than 10 shipments	56.90%	68.47%
(6)	Share of (5) that are active in quarters (−8) through (+7)	76.54%	71.53%
(7)	Share of (6) with controls	79.16%	85.50%
	Share of (7) with a single match in step (4)	19.69%	21.84%
	Share of (7) with multiple matches in step (4)	80.31%	86.49%

Notes: Steps in the creation of analysis dataset. For each step, we show the share of recalls and U.S. importing firms we retain. "HS6" denotes an imported product, "country" the source country, "MID" the abbreviated foreign supplier name, and "date" the shipment date. In most of our analyses, we restrict the sample to "active" U.S. firms, with at least one imported shipment per quarter in quarters (−8) through (+7); Figure B-2 relaxes this condition. Step (7) forms our baseline analysis sample. The last two rows show the number of recalls and firms associated with single or multiple shipments. Firm shares in the last two rows do not add up to 100% because a single firm may experience multiple recalls.
Source: Authors' calculations from CPSC, BR, and LFTTD data.

Table A-2. Supplementary Summary Statistics

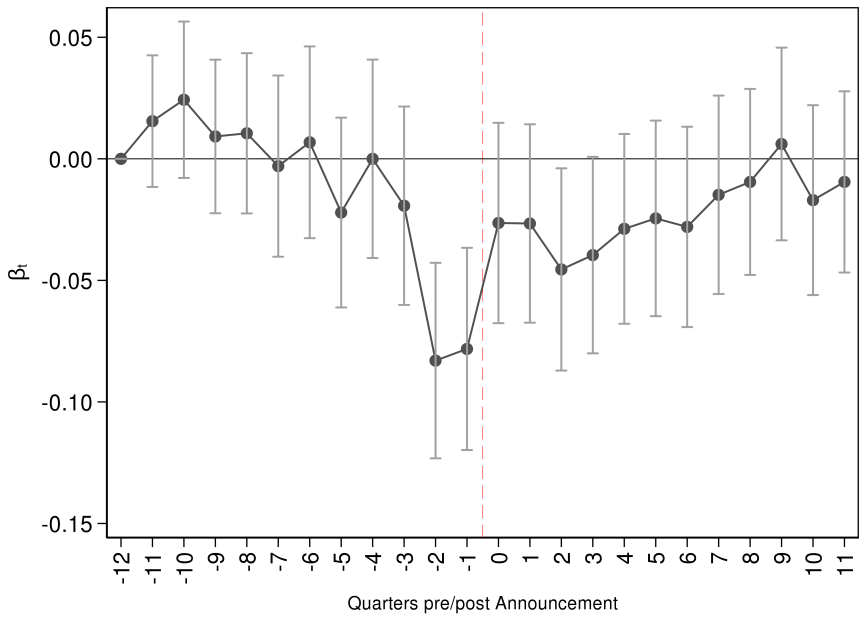
	Treatment	Control
A. Buyer Sample[a]		
Supplier share of buyer's import value	0.1160	0.0991
	(0.2100)	(0.1882)
Supplier share of buyer's import value for HS6 product	0.3206	0.2465
	(0.3560)	(0.3326)
HS6 share of pair's trade value	0.3966	0.3584
	(0.3993)	(0.3895)
Number of suppliers	337.1	465.7
	(2,381)	(3,328)
Number of suppliers for HS6 product	6.700	7.960
	(17.85)	(16.28)
Number of HS6 products purchased	95.47	91.77
	(217.40)	(308.10)
Number of source countries	11.92	10.75
	(17.90)	(15.32)
Years of trade[b]	6.812	6.319
	(4.895)	(4.617)
B. Supplier Sample		
Number of HS6 products sold	50.37	55.32
	(73.26)	(57.99)

Notes: Mean, with standard deviation in parentheses, for selected variables in quarters $\{-12, -11, -10, -9, -8, -7, -6, -5\}$ relative to the recall announcement, by treatment and control groups. All numbers have been rounded to comply with Census Bureau disclosure-avoidance rules.

[a] Based on 750 treated relationships and 4,200 control relationships per quarter.

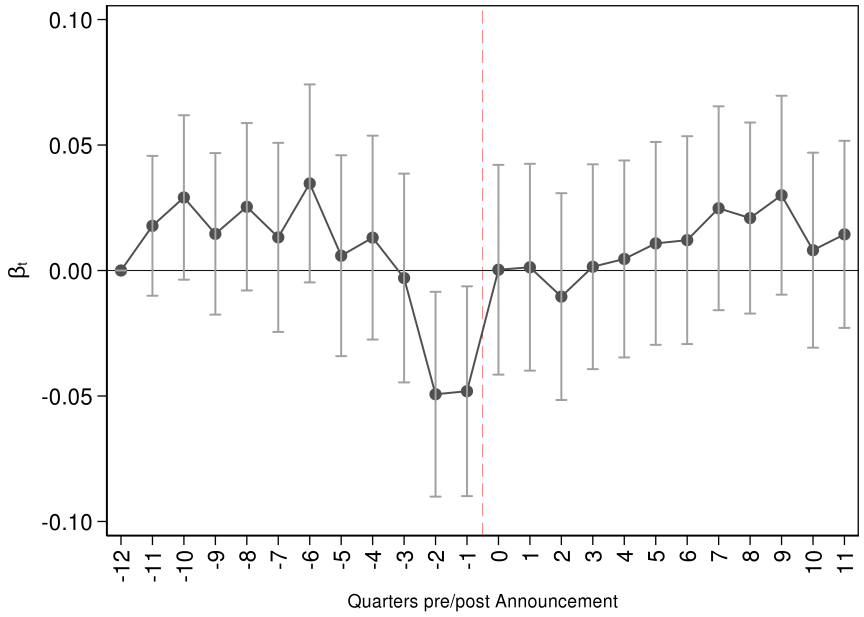
[b] The number of years a buyer-supplier pair transacts between 1992 and the first recall.

Figure B-1. Extensive-Margin Dynamics: Buyer-Supplier-Product, Expanded Control Group with Indirect Traders



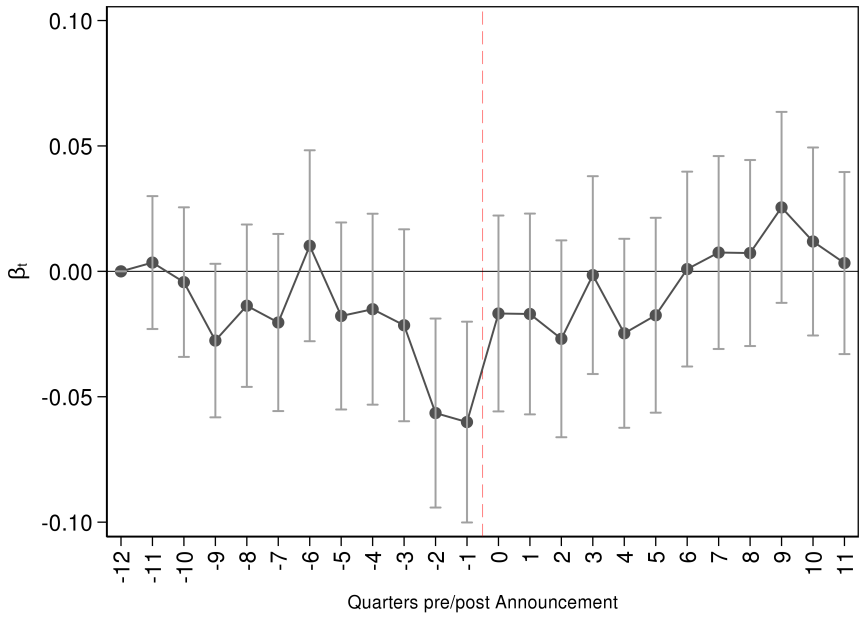
Notes: Coefficient estimates and 95% confidence intervals showing the probability of trade between treated trading pairs in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control trading pairs. The regression includes quarter and buyer-supplier fixed effects. Controls are expanded to include those trading in the same HS6 product as the recalled product, as well as control firms that have had a trading relationship with treated firms. Standard errors are clustered by recall.

Figure B-2. Extensive-Margin Dynamics: Buyer-Supplier-Product, Expanded Control Group with Inactive Firms



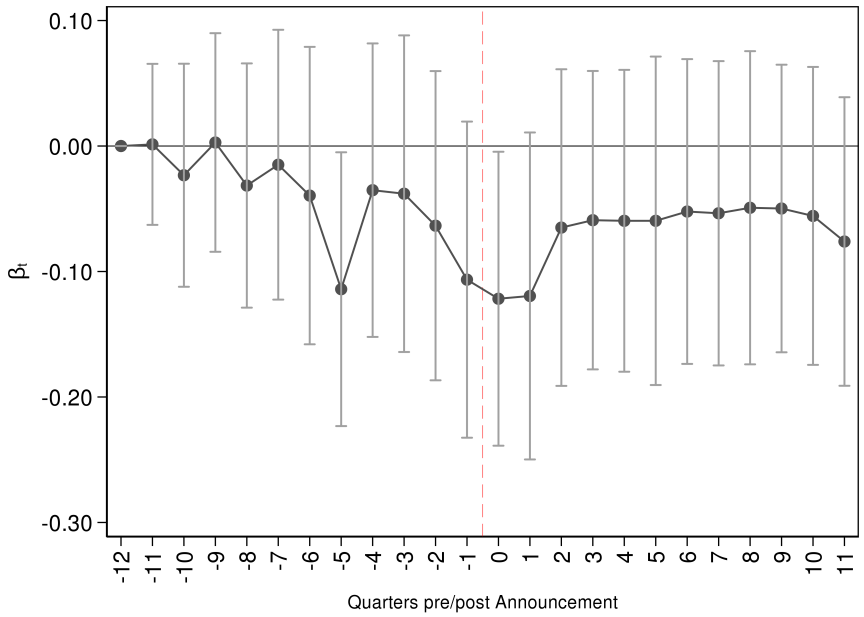
Notes: Coefficient estimates and 95% confidence intervals showing the probability of trade between the treated trading pairs in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control trading pairs. The regression includes quarter and buyer-supplier fixed effects. Both treatment and controls are expanded to include firms that have no trade transactions in quarters (−8) through (+7). Standard errors are clustered by recall.

Figure B-3. Extensive-Margin Dynamics: Buyer-Supplier-Product, Supplier-Based Controls



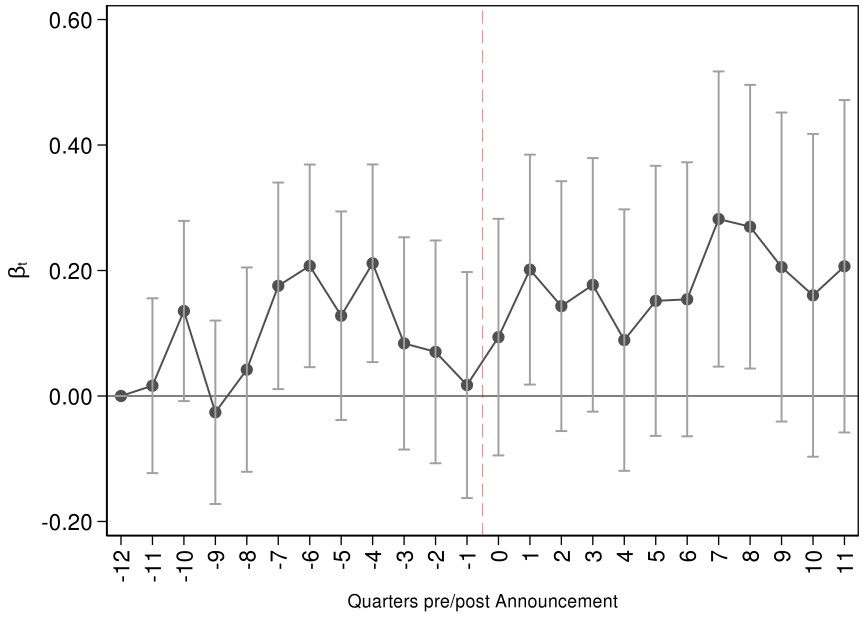
Notes: Coefficient estimates and 95% confidence intervals showing the probability of trade between the treated trading pairs in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control trading pairs. The regression includes quarter and buyer-supplier fixed effects. Controls are selected based on supplier, rather than buyer, characteristics. Standard errors are clustered by recall.

Figure B-4. Extensive-Margin Dynamics: Buyer-Supplier-Product, Same Firms in Treatment and Control Groups



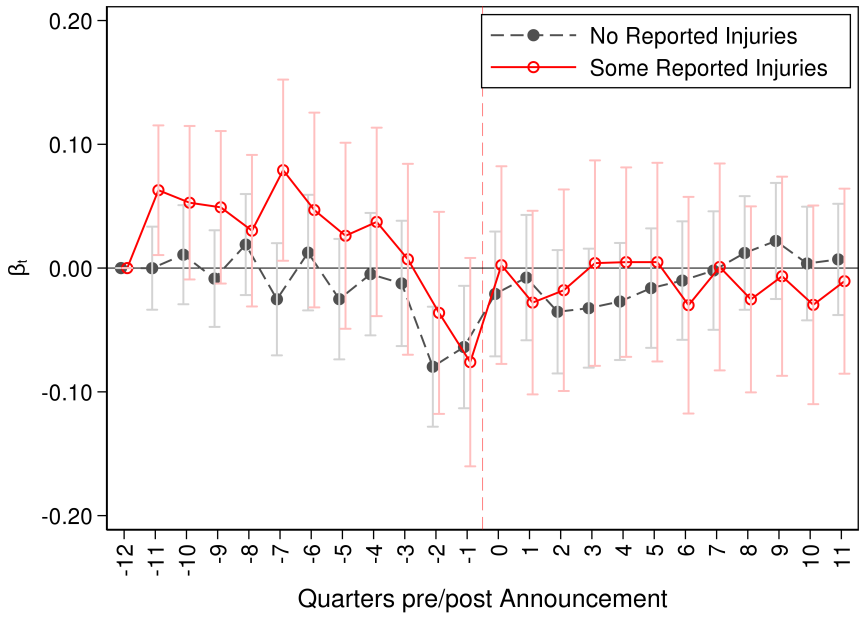
Notes: Coefficient estimates and 95% confidence intervals showing the probability of trade between the treated trading pairs in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control trading pairs. The regression includes quarter and buyer-supplier fixed effects. Control relationships consist of the same U.S. firms as treated relationships, two years prior to the recall.

Figure B-5. Intensive-Margin Dynamics: Buyer-Supplier



Notes: Coefficient estimates and 95% confidence intervals showing the log value of trade between the treated trading pairs, conditional on having any trade in a given quarter, for 12 quarters before and after a public recall announcement, relative to control trading pairs. The regression includes quarter and buyer-supplier fixed effects. Standard errors are clustered by recall.

Figure B-6. Extensive-Margin Dynamics: Buyer-Supplier-Product, Heterogeneous Effects by Injury Report



Notes: Coefficient estimates and 95% confidence intervals showing the probability of trade between the treated trading pairs in the HS6 product category, for 12 quarters before and after a public recall announcement, relative to control trading pairs, interacted with an indicator for the recall notice containing injury information. The regression includes quarter and buyer-supplier fixed effects. Standard errors are clustered by recall.

Table B-1. Predicting Product Recalls

Probability of Recall × 100,000				
	(1)	(2)	(3)	(4)
Buyer's prior recalls	0.001			0.005
(1992-2006)	(0.049)			(0.052)
Supplier's prior recalls		0.095		0.189
(1992-2006)		(0.475)		(0.349)
Relationship's prior recalls			0.947	1.148
(1992-2006)			(1.753)	(1.848)
Buyer's value of shipments	✓			✓
Supplier's value of shipments		✓		✓
Relationship's value of shipments			✓	✓

Notes: Sample includes LFTTD shipments in 2007 aggregated to the buyer-supplier-HS10-date level. The dependent variable is 100,000 times the assigned probability that the shipment was recalled. Standard errors are clustered by HS10. Each regression has 13,410,000 observations. All regressions control for HS10 and country-of-origin fixed effects. Value of shipments controls refer to logged value accumulated over the 1992-2006 period.
* significant at 10%; ** significant at 5%; *** significant at 1%

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